HKE's PDA COLLEGE OF ENGINEERING (AUTONOMOUS), KALABURAGI- KARNATAKA

Department of Electronics and Communication Engineering

[**Inspire** : **" Don't allow the fear of failure, to override the joy of learning "**]

Course Name: Introduction to Electronics and Communication Engineering(IEC)

Course Code	**KKIEC104C/204C**	Semester	I/II
Teaching hours per week (L:T:P:S)	3:0:0:0	CIE Marks	50
Total Hours	42 Hours (Theory)	SEE Marks	50
Credits	3	Exam Hours	03
Examination Type (SEE)	Theory		

Course Objectives:

- To study the power supplies and its important components
- To understand the concepts of amplifiers, oscillators, operational amplifier and its applications
- To study the basics of analog and digital communication systems
- To study the fundamentals of embedded systems and digital logic gates

Module-1	Teaching Hours
Power Supply: Block Diagram of power supply, Analysis of Rectifiers: Half wave and Full wave rectifiers with capacitive filter, Zener voltage regulator, Voltage multiplier (Doublers). **Amplifiers**: Types of Amplifiers, Gain, Input and Output Resistance of CE transistor amplifier, Frequency Response of two stages RC coupled amplifier. Text1:PageNo:117-128,139-146	**09**
Module-2	
Oscillators: Concept of feedback: Positive and Negative Feedback, Condition for Oscillations, RC Phase shift oscillator, Hartley and Colpitt's oscillator, Crystal Controlled Oscillators. (Frequency expressions: No derivation) **Operational Amplifiers:** Basic block diagram, Ideal Characteristics, Parameters: Gain, Input and Output Impedance, CMRR, Bandwidth, Slew rate, PSRR, Input offset voltage and Input offset current, Concept of virtual ground. Op amp Applications**:** Inverting and Non inverting amplifiers, voltage follower, adder, comparator, integrator and differentiator. Text1:PageNo:179-186,165-169,171-175	**09**
Module-3	
Communication Systems: Introduction, Frequency Spectrum, Modern Communication System Scheme: Information Source, Input transducer, Transmitter, Channel, Noise, Receiver, Concepts of Radio wave Propagation (Ground, space and sky wave propagation) **Modulation Schemes:** Needs of modulation, Analog modulation: Analysis of AM, FM (only waveform and expression), Digital modulation: ASK, FSK and PSK (Definition and waveforms), Advantage of Digital communication over Analog communication, Multiplexing: FDM and TDM Text2:1.1,1.2,1.3,1.4,1.5,1.6,1.9,1.12,1.15,2.2.1,3.2.1,6.1,6.11,6.12,6.13,6.15,6.16	**08**
Module-4	
Embedded Systems: Definition, Embedded Systems Vs General Computing Systems, Classification of Embedded Systems, Major Application Areas of Embedded Systems, Purpose of an Embedded System, Core of The Embedded System: Microprocessors, GPP Vs ASIP, Microcontrollers, Microprocessor Vs Microcontroller, DSP, RISC V/S CISC, Memory: RAM and ROM, Sensors, Actuators, LED, 7-Segment LED Display. Text3:1.1,1.2,1.4,1.5,1.6,2.1.1.1-2.1.1.6,2.2.1,2.3.1,2.3.2,2.3.3.1,2.3.3.2.	**08**
Modules-5	
Boolean Algebra and Logic Circuits: Binary Numbers, Number Base Conversion: Binary, Decimal, Octal and Hexa Decimal Numbers and Vice-Versa, 1's and 2's complements, Arithmetic operations using 2's complement method, Boolean Algebra: Properties and DeMorgans Theorem, Boolean Functions, Canonical and Standard Forms, Digital Logic Gates: Basic and Universal gates. **Combinational Logic:** Introduction, Design of Adders: Half Adder, Full Adder. Text4:1.2,1.3,1.4,1.5,2.1,2.3,2.4,2.5,2.7,4.1,4.2,4.3.	**08**

Question paper pattern:

- The question paper shall have five modules for 100 marks;
- Each full question carries 20 marks.
- Two questions to be set in each module (total ten questions).
- The candidate will have to answer one full question from each module.

Note: There can be a maximum of 4 subsections in each Question.

Text books:

1. Mike Tooley "Electronic Circuits Fundamentals & Applications," 5th Edition, Elsevier, 2020.
2. S L Kakani and Priyanka Punglia, 'Communication Systems', Ist Edition, New Age International Publisher, 2017.
3. K V Shibu, 'Introduction to Embedded Systems', 2nd Edition, McGraw Hill Education (India), Private Limited, 2019.
4. Digital Logic and Computer Design, M. Morris Mano, Pearson Education, 2017, ISBN-978-93-325-4252-5.

E books and online course materials:

- https://nptel.ac.in/courses/122106025
- https://nptel.ac.in/courses/108105132

Course outcomes: On completion of the course, the student will have the ability to,

Course Code	CO #	Course Outcome (CO)
KKIEC104C/204C	CO1	Analyze basic electronic circuits using the principles of rectifiers, voltage regulators, and amplifiers.
	CO2	Analyze the behavior of analog circuits including oscillators and operational amplifiers in signal generation and conditioning applications.
	CO3	Illustrate the fundamental concepts of analog and digital modulation techniques based on their characteristics and suitability for communication systems.
	CO4	Interpret the structure and functionality of embedded systems and digital logic components such as microcontrollers, sensors, and logic gates.
	CO5	Apply number system conversions and Boolean algebra to design and implement basic combinational logic circuits.

Module-1: Power Supply and Amplifiers

- The block diagram of a d.c. power supply is shown in Fig-1.1

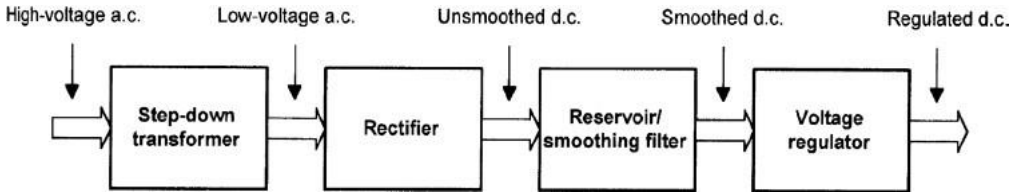


Figure-1.1 Block diagram of a d.c. power supply

- The mains input is relatively high voltage, a step-down transformer of appropriate turns ratio is used to convert this to allow voltage.
- The a.c. output from the transformer secondary is then rectified using conventional silicon rectifier diodes to produce an unsmoothed (or **pulsating d.c.**) output.
- This is then smoothed and filtered before being applied to a circuit which will **regulate** (or **stabilize**) the output voltage so that it remains relatively constant in spite of variations in both load current and incoming mains voltage.
- Fig. 1.2 shows the realization of the block diagram of a d.c. power supply using the electronic components in Fig. 1.1

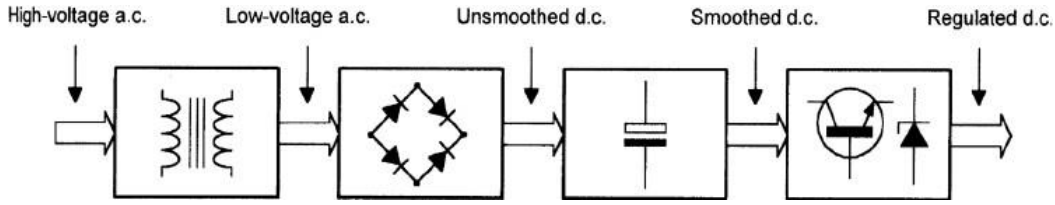


Figure 1.2 Block diagram of a d.c. power supply showing principal components

- The iron-cored step-down transformer feeds a rectifier arrangement (often based ona bridge circuit).
- The output of the rectifier is then applied to a high-value **reservoir** capacitor. This capacitor stores a considerable amount of charge and is being constantly topped-up by the rectifier arrangement. The capacitor also helps to smooth out the voltage pulses produced by the rectifier.
- Finally, a stabilizing circuit (often based on a **series transistor regulator** and a zener diode **voltage reference**) provides a constant output voltage. .

1. Rectifiers

- Rectifiers are the circuits which converts a.c voltage to pulsating d.c voltage. Rectifiers can be grouped into two types:
  1. Half-wave Rectifier
  2. Full-wave Rectifier
- The simplest form of rectifier circuit makes use of a single diode and, since it operates on only either positive or negative half-cycles of the supply, it is known as a half-wave rectifier. Fig. 1.3 shows a simple half- wave rectifier circuit.

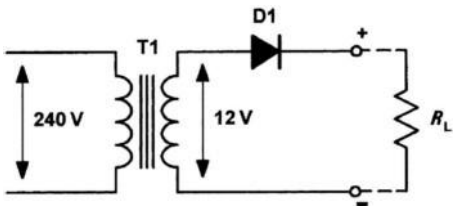


Figure 1.3 A simple half-wave rectifier circuit

- The mains voltage (220 to 240V) is applied to the primary of a step-down transformer (T1). The secondary of T1 steps down the 240 V r.m.s. to 12 V r.m.s.
- Diode D1 will only allow the current to flow in the direction shown (i.e. from cathode to anode). D1 will be forward biased during each positive half-cycle and will effectively behave like a closed switch as shown in Fig. 1.4.

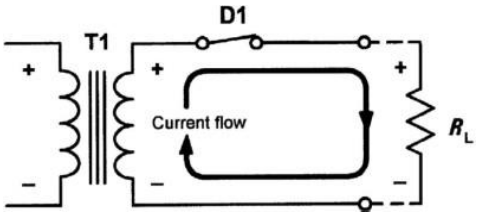


Figure 1.4 Half-wave rectifier circuit with D1 conducting (positive-going half- cycles of secondary voltage)

- When the circuit current tries to flow in the opposite direction, the voltage bias across the diode will be reversed, causing the diode to act like an open switch as shown in Fig. 1.5.

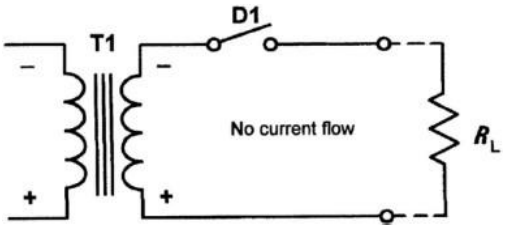


Figure 1.5 half-wave rectifier with D1 not conducting (negative-going half- cycles of secondary voltage)

- During positive half cycle, the diode D1 is forward biased, thus the current flows through the load R_L and voltage is developed across it.
- During negative half cycle, the diode D1 is reverse biased, thus there will be no flow of current through the load RL, thereby the output voltage is zero.
- The input and output voltage waveform of a half-wave rectifier is shown in Fig. 1.6.

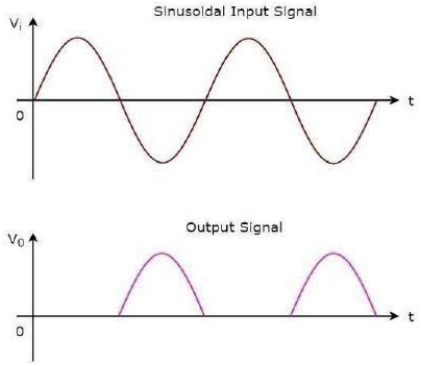


Figure 1.6 The input and output voltage waveform of a half-wave rectifier

- The relation between turns ratio and voltage of the primary and secondary of the transformer is given by:

$$\frac{N_1}{N_2} = \frac{V_{p(rms)}}{V_{s(rms)}}$$

- The peak voltage output from the transformer's secondary winding will

$$V_{pk} = 1.414\, V_{s(rms)}$$

1.1. Half-Wave Rectifier:

- A half wave rectifier is defined as a type of rectifier that only allows one half-cycle of an AC voltage waveform to pass, blocking the other half-cycle.
- Half-wave rectifiers are used to convert AC voltage to DC voltage, and only require a single diode to construct as in fig-1.

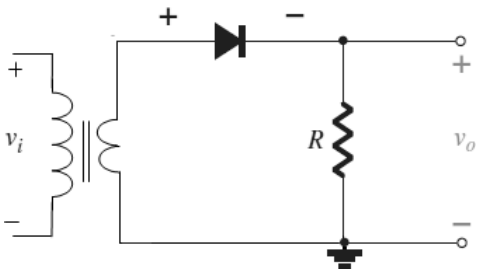
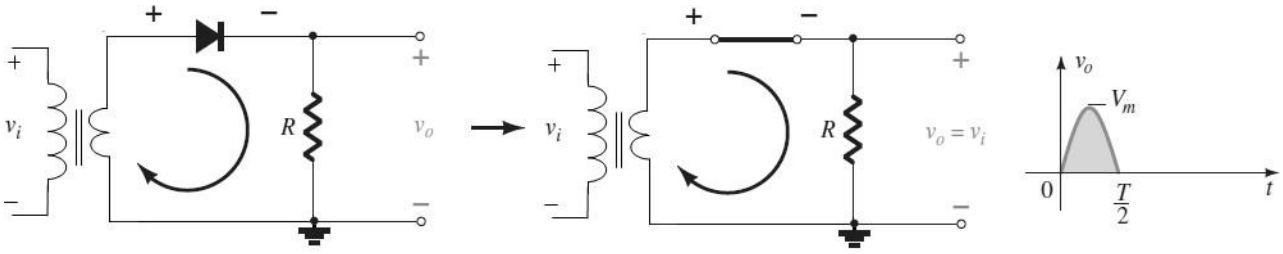


Fig-1.1

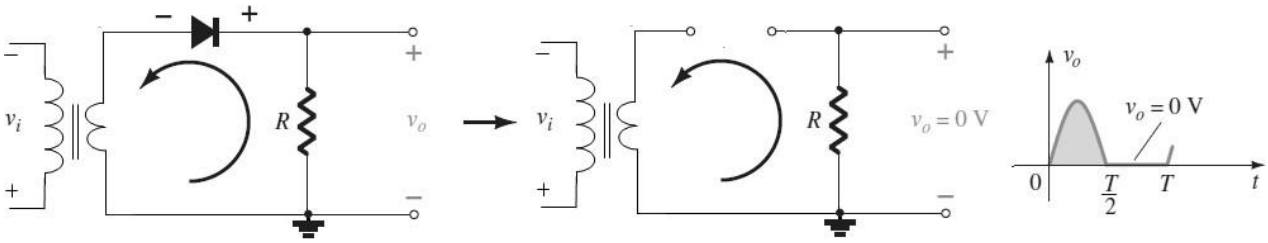
- ***Operation:***
  - During positive half cycle of the AC sine wave, the diode is *forward biased* as the anode is positive with respect to the cathode resulting in current flowing through the diode.
  - Since the load is resistive, the current flowing in the load resistor is therefore proportional to the voltage, and the voltage across the load resistor will therefore be the same as the supply voltage, V_i (minus V_f(threshold voltage)), that is the "DC" voltage across the load is sinusoidal for the first half cycle only so $v_o = v_i$as in fig-1.2



Conduction region ($0 \rightarrow T/2$).

Fig-1.2

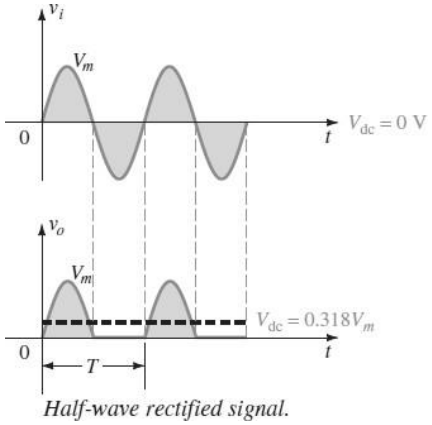
  - During negative half cycle of the AC sinusoidal input waveform, the diode is *reverse biased* as the anode is negative with respect to the cathode. Therefore, no current flows through the diode or circuit. Then in the negative half cycle of the supply, no current flows in the load resistor as no voltage appears across it therefore, $v_o = 0$ as in



Nonconduction region ($T/2 \rightarrow T$).

Fig-1.2

  - The input $v_i$and the output $v_o$are sketched together in below fig-1.3



Half-wave rectified signal.

Fig-1.3

Average value of Half Wave rectifier:

$$V_{dc} = \frac{1}{T}\int_0^1 V\,dt = \frac{1}{T}\int_0 v_m sin\omega t\,dt = \frac{1}{T}\int_0^{/2} v_m\, sin\omega t\,dt + \int_{T/2}^{T} 0\;dt = \frac{v}{T}\left\{\frac{-cos\omega t}{\omega}\right\}_0^{T/2}$$

$$V_{dc} = \frac{v_m}{\omega T}\{-cos(\frac{2\pi}{T}\times\frac{T}{2}) + cos0\} = \frac{v_m}{T}\times\frac{T}{2\pi}\{-(\pi) + cos0\}$$

$$V_{dc} = \frac{-}{2\pi T}\{-co(\pi) + cos0\} = \frac{v_m}{2\pi}\{1+1\} \rightarrow \boldsymbol{V_{dc} = \frac{v_m}{\pi} = 0.3183 v_m}$$

RMS value of Half Wave rectifier

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^T V^2\,dt} \rightarrow V^2_{rms} = \frac{1}{T}\int_0^T V^2\,dt = \frac{1}{T}\int_0^T V^2\,dt = \frac{1}{T}\int_0^{T/2} v^2_m\, sin^2\omega t\,dt$$

$$V^2_{rms} = \frac{v^2_m}{T}\int_0^{T/2} sin^2\omega t\,dt = \frac{v^2_m}{T}\int_0^{T/2}\frac{(1-cos2\omega t)}{2}\,dt = \frac{v^2_m}{2T}(t - \frac{sin2\omega t}{2\omega})_0^{T/2}$$

$$V^2_{rms} = \frac{v^2_m}{2T}\{\frac{T}{2} - \frac{sin(\frac{4\pi}{T}\times\frac{T}{2})}{2\omega} - 0 + \frac{(0)}{2\omega}\} = \frac{v^2_m}{2T}\times\frac{T}{2} = \frac{v^2_m}{4} \rightarrow \boldsymbol{V_{rms} = \frac{v_m}{2}}$$

Ripple Factor of Half Wave rectifier

The measure of ripples (pulsating components) present in the output is obtained by a factor called ripple factor denoted by γ. Smaller the ripple factor closer is the output toa pure dc.∴ For HWR, γ is

$$\gamma = \sqrt{(\frac{V_{rms}}{V_{avg}})^2 - 1} = \sqrt{(\frac{v_m/2}{v_m/\pi})^2 - 1} \rightarrow \boldsymbol{\gamma = 1.21}$$

Peak Inverse Voltage (PIV):

It is the peak voltage across the diode in reverse biased condition when diode is not conducting.

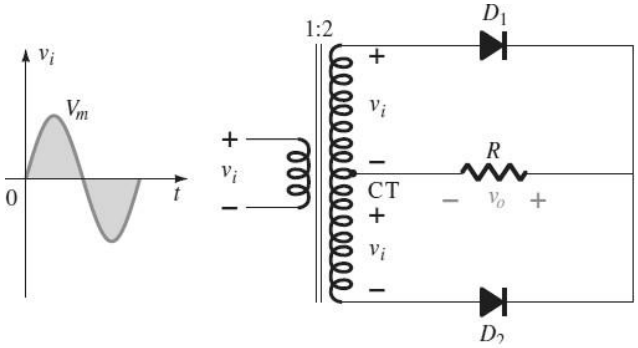
∴For HWR, PIV =$\boldsymbol{V_m}$

1.2. FullWave Rectifier:

- A full wave rectifier is a type of rectifier which converts both half cycles of the AC signal into pulsating DC signal. The full wave rectifier is further classified into two types: Center Tapped Full Wave Rectifier and Full Wave Bridge Rectifier.

1.2.1. Center Tapped Full Wave Rectifier (Bi-phase rectifier)

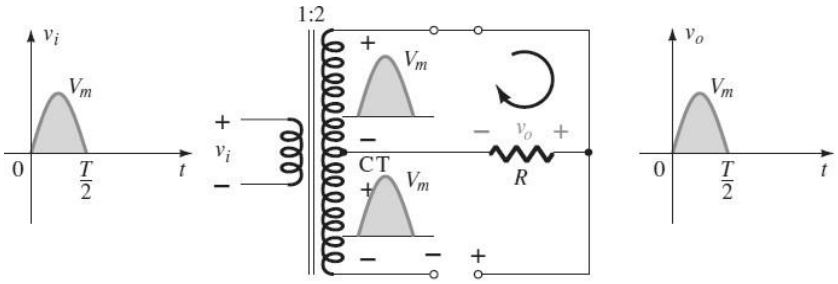
- A center tapped full wave rectifier is a type of rectifier which uses a center tapped transformer and two diodes to convert the complete AC signal into DC signal.



Center-tapped transformer full-wave rectifier.

Fig-1.2.1.1

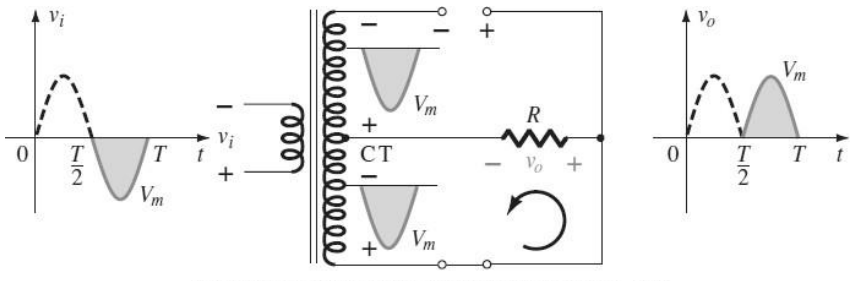
- During the positive half cycle, current flows only in the upper part of the circuit while the lower part of the circuit carry no current to the load because the diode D_2 is reverse biased as in below fig. Thus, during the positive half cycle of the input AC signal, only diode D_1 allows electric current while diode D_2 does not allow electric current.



Network conditions for the positive region of v_i.

Fig-1.2.1.2

- During the negative half cycle, current flows only in the lower part of the circuit while the upper part of the circuit carry no current to the load because the diode D_1 is reverse biasedas in below fig. Thus, during the negative half cycle of the input AC signal, only diode D_2 allows electric current while diode D_1 does not allow electric current.



Network conditions for the negative region of v_i.

Fig-1.2.1.3

- Thus, the diode D_1 allows electric current during the positive half cycle and diode D_2 allows electric current during the negative half cycle of the input AC signal. As a

result, both half cycles (positive and negative) of the input AC signal are allowed. So the output DC voltage is almost equal to the input AC voltage.

1.2.2. Full Wave Bridge Rectifier

- This type of single phase rectifier uses four individual rectifying diodes connected in a closed loop "bridge" configuration to produce the desired output as shown in below fig-11.2.1

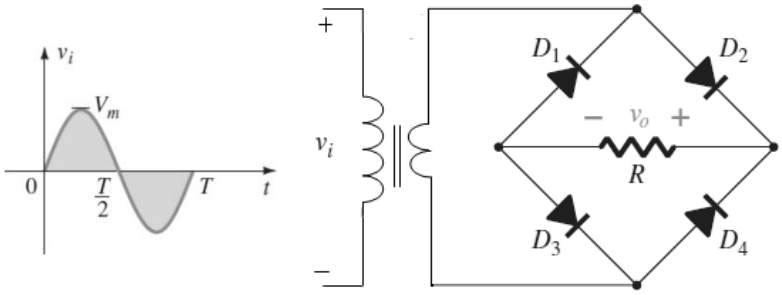
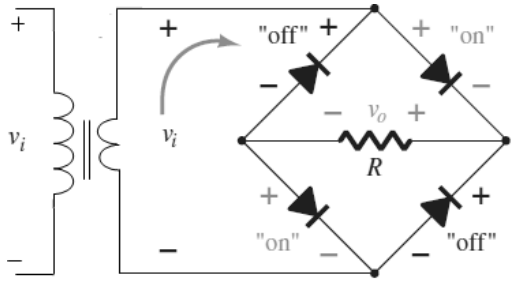
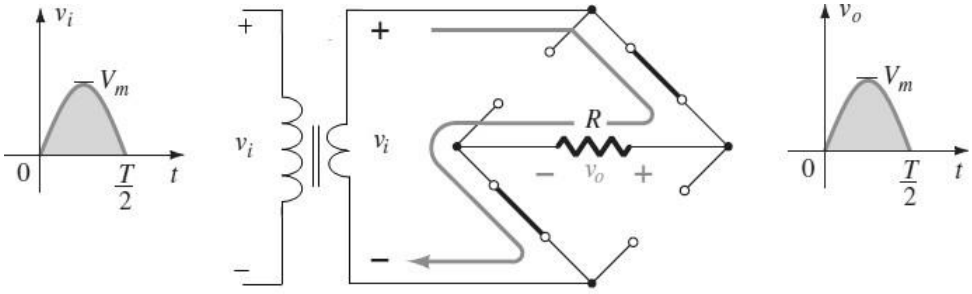


Fig-1.2.2.1

- During the period t =0 to $T/2$ the polarity of the input is as shown in below Fig-1.2.2.2. The resulting polarities across the ideal diodes are also shown. Diodes $D2$ and $D3$ are conducting, whereas $D1$ and $D4$ are in the "off" state. The net result is the configuration as in of fig with its indicated current and polarity across R. Since the diodes are ideal, the load voltage is$v_o = v_i$, as shown in the same figure.



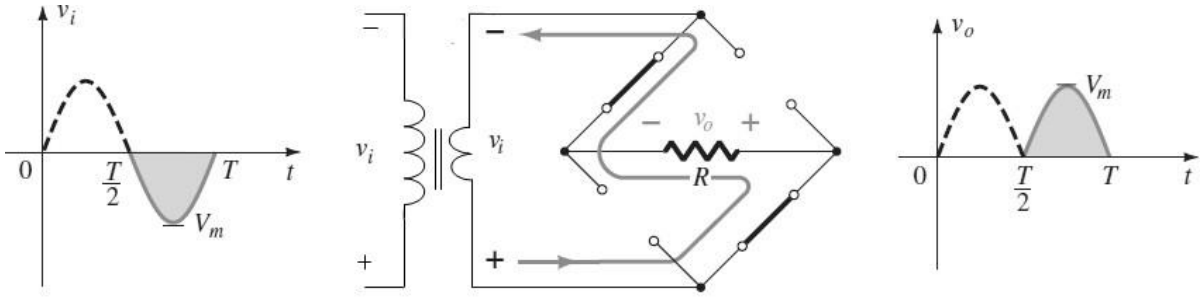
Network for the period
$0 \rightarrow T/2$ of the input voltage v_i.



Conduction path for the positive region of v_i.

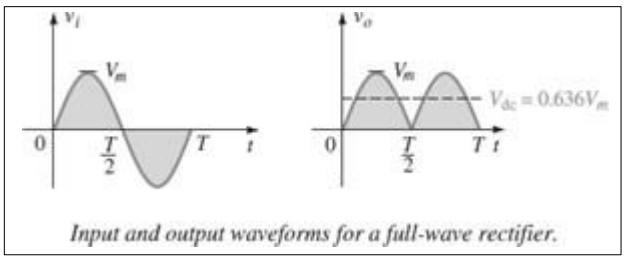
Fig-1.2.2.2

- For the negative region of the input the conducting diodes are $D1$ and $D4$, resulting in the configuration of fig-1.2.2.3. The important result is that the polarity across the load resistor R is the same as in fig-1.2.2.2, establishing a second positive pulse, as shown in fig-1.2.2.4. Overone full cycle the input and output voltages will appear as shown in fig-1.2.2.4.



Conduction path for the negative region of v_i.

Fig-1.2.2.3



Input and output waveforms for a full-wave rectifier.

Fig-1.2.2.4

Average value of Full Wave rectifier

$$V_{dc} = \frac{1}{T}\int_0^T V\,dt = \frac{1}{T}\int_0^T v_m \sin\omega t\,dt = \frac{1}{T}2\int_0^{\frac{T}{2}} v_m \sin\omega t\,dt = \frac{2v_m}{T}\left\{\frac{-\cos\omega t}{\omega}\right\}_0^{\frac{T}{2}}$$

$$V_{dc} = \frac{-2v_m}{\omega T}\left\{\cos\left(\frac{2\pi}{T}\times\frac{T}{2}\right) - \cos 0\right\} = \frac{-2v_m}{\omega T}\{(\pi) - \cos 0\} = \frac{-2v_m}{T}\times\frac{T}{2\pi}\{-1-1\}$$

$$V_{dc} = \frac{2v_m}{\pi} \rightarrow \boldsymbol{V_{dc} = 0.636 v_m}$$

RMS value of Full Wave rectifier

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^{T/2} V^2\,dt} = \frac{2}{T}\int_0^{T/2} V^2\,dt = \frac{2}{T}\int_0^{T/2} v^2_m \sin^2\omega t\,dt = \frac{2v^2_m}{T}\int_0^{T/2} \sin^2\omega t\,dt$$

$$V^2_{rms} = \frac{2v^2_m}{T}\int_0^{T/2} \frac{(1-\cos 2\omega t)}{2}\,dt = \frac{v^2_m}{T}\left(t - \frac{\sin 2\omega t}{2\omega}\right)_0^{T/2}$$

$$V^2_{rms} = \frac{v^2}{T}\left\{-\frac{\sin\left(\frac{4\pi}{T}\times\frac{T}{2}\right)}{2\omega} - 0 + \frac{(0)}{2\omega}\right\} = \frac{v^2_m}{T}\times\frac{T}{2} = \frac{v^2_m}{2} \rightarrow \boldsymbol{V_{rms} = \frac{v_m}{\sqrt{2}}}$$

Ripple Factor of Full Wave rectifier

The measure of ripples (pulsating components) present in the output is obtained by a factor called ripple factor denoted byγ. Smaller the ripple factor closer is the output toa pure dc.

$\therefore$ For FWR, γ is

$$\gamma = \sqrt{\left(\frac{V_{rms}}{V_{avg}}\right)^2 - 1} = \sqrt{\left(\frac{v_m/\sqrt{2}}{2v_m/\pi}\right)^2 - 1} \rightarrow \boldsymbol{\gamma = 0.48}$$

Peak Inverse Voltage (PIV):

It is the peak voltage across the diode in reverse biased condition when diode is not conducting.

∴For FWR, PIV =$\mathbf{2V_m}$

Other useful Formulae:

1. Power consumed by the load or dc output power $\mathbf{P_{dc} = I_{dc}^2 \times R_L}$

$$= \frac{\mathbf{V_{dc}}^2}{\mathbf{R_L}} \quad Where,\ I_{dc} = \frac{V_{dc}}{R_L}$$

2. Power input to the load or AC input power $\mathbf{P_{ac} = I_{rms}^2 \times (r_f + R_L)}$

$$= \frac{\mathbf{V_{rms}}^2}{(\mathbf{r_f + R_L})}$$

$$Where,\ I_{rms} = \frac{V_{rms}}{R_L}\ \&\ r_f\ is\ forward\ resistance\ of\ diode$$

3. Rectifier efficiency is defined as the ratio of the DC output power to the AC input power.

$$\textbf{Efficiency}, \eta\ \% = \frac{\textbf{dc output power}}{\textbf{ac input power}} = \frac{\mathbf{P_{dc}}}{\mathbf{P_{ac}}}$$

2. Reservoir and smoothing circuits

- Fig. 2.1 shows a simple half-wave rectifier circuit with reservoir capacitor.

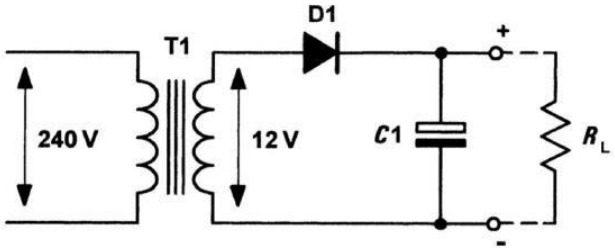


Fig-2.1A simple half-wave rectifier circuit with reservoir capacitor

- The capacitor, C1, has been added to ensure that the output voltage remains at, or near, the peak voltage even when the diode is not conducting.
- When the primary voltage is first applied to T1, the first positive half- cycle output from the secondary will charge C1 to the peak value seen across R_L.
- Hence C1 charges to 16.3 V at the peak of the positive half-cycle. Because C1 and R_L are in parallel, the voltage across R_L will be the same as that across C1.
- The time required for C1 to charge to the maximum (peak) level is determined by the charging circuit time constant (the series resistance multiplied by the capacitance value).
- In this circuit, the series resistance comprises the secondary winding resistance together with the forward resistance of the diode and the (minimal) resistance of the wiring and connections. Hence C1 charges very rapidly as soon as D1 starts to conduct.
- The time required for C1 to discharge is, in contrast, very much greater. The discharge time constant is determined by the capacitance value and the load resistance, R_L. In practice, R_L is very much larger than the resistance of the secondary circuit and hence C1

takes an appreciable time to discharge. During this time, D1 will be reverse biased and will thus be held in its non-conducting state.

- As a consequence, the only discharge path for C1 is through R_L. C1 is referred to as a reservoir capacitor. It stores charge during the positive half-cycles of secondary voltage and releases it during the negative half- cycles.
- The circuit of Fig-2.1 is thus able to maintain a reasonably constant output voltage across R_L. Even so, C1 will discharge by a small amount during the negative half-cycle periods from the transformer secondary.
- Fig-2.2 shows the secondary voltage waveform together with the voltage developed across R_L with and without C1 present.

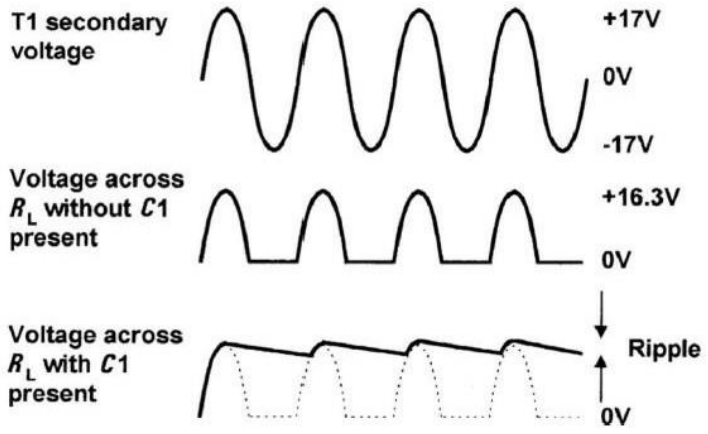


Fig-2.2: A simple half-wave rectifier circuit with reservoir capacitor

- This gives rise to a small variation in the d.c. output voltage (known as ripple). Since ripple is undesirable an additional precaution has to be taken to reduce it.
- One obvious method of reducing the amplitude of the ripple is that of simply increasing the discharge time constant. This can be achieved either by increasing the value of C1 or by increasing the resistance value of R_L.
- Fig-2.4 shows a further refinement of the simple power supply circuit.

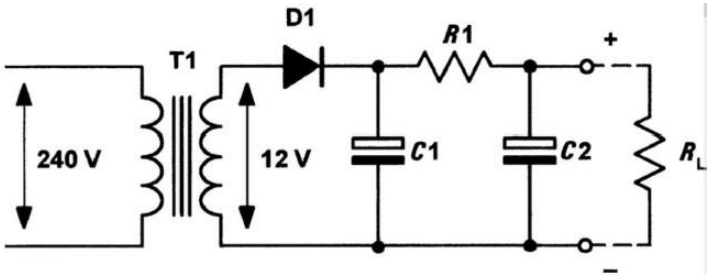


Fig-2.4: Half-wave rectifier circuit with R–C smoothing filter

- This circuit employs two additional components, R1 and C1, which act as a filter to remove the ripple.
- The value of C1 is chosen so that the component exhibits a negligible reactance at the ripple frequency (50 Hz for a half-wave rectifier or 100 Hz for a full-wave rectifier).
- In effect, R1 and C1 act like a potential divider. The amount of ripple is reduced by an approximate factor equal to:

$$\frac{X_C}{\sqrt{R^2 + X_C^2}}$$

3. Voltage regulators

- Voltage regulator is a circuit that maintains a constant d.c output voltage irrespective of variations in the input line voltage or in the load.
- Voltage regulator is one of the important applications of a Zener diode.
- A simple voltage regulator is shown in Fig-3.1

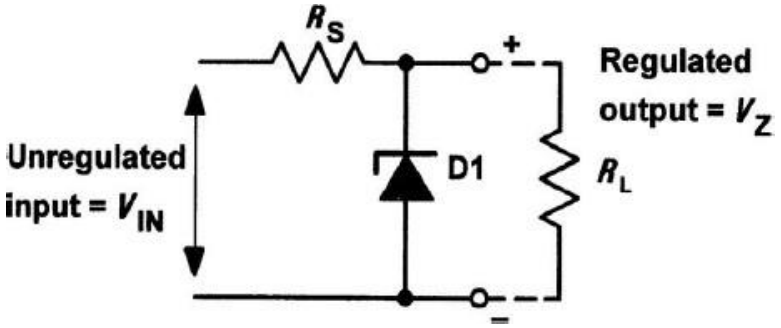


Fig-3.1: A simple shunt Zener voltage regulator

- R_S is included to limit the zener current to a safe value when the load is disconnected.
- When a load (R_L) is connected, the zener current (I_Z) will fall as current is diverted into the load resistance (it is usual to allow a minimum current of 2 mA to 5 mA in order to ensure that the diode regulates).
- The output voltage (V_Z) will remain at the Zener voltage until regulation fails at the point at which the potential divider formed by R_S and R_L produces a lower output voltage that is less than V_Z.
- The ratio of R_S to R_L is thus important. Regulated output VZ is given by:

$$V_Z = V_{IN} \times \frac{R_L}{R_L + R_S}$$

where V_{IN} is the unregulated input voltage.

4. <u>Output resistance and voltage regulation</u>

- In a perfect power supply, the output voltage would remain constant regardless of the current taken by the load.
- Output resistance Rout is defined as the change in output voltage divided by the corresponding change in output current and hence is given by

$$R_{out} = \frac{change\ in\ output\ voltage}{change\ in\ output\ current} = \frac{\Delta V_{out}}{\Delta I_{out}}$$

where$\Delta Iout$ represents a small change in output (load) current and $\Delta Vout$represents a corresponding small change in output voltage.

- The regulation of a power supply is given by the relationship:

$$Regulation = \frac{change\ in\ output\ voltage}{change\ in\ line\ (iutput)\ voltage} \times 100$$

- Ideally, the value of regulation should be very small. Simple shunt zener diode regulators of the type shown in Fig. 1.17 are capable of producing values of regulation of 5% to 10%.
- More sophisticated circuits based on discrete components produce values of between 1% and 5% and integrated circuit regulators often provide values of 1% or less.

<u>Voltage multipliers</u>

- By adding a second diode and capacitor, the output of the simple half- wave rectifier can be increased. A voltage doubler using this technique is shown in Fig. 1.19.

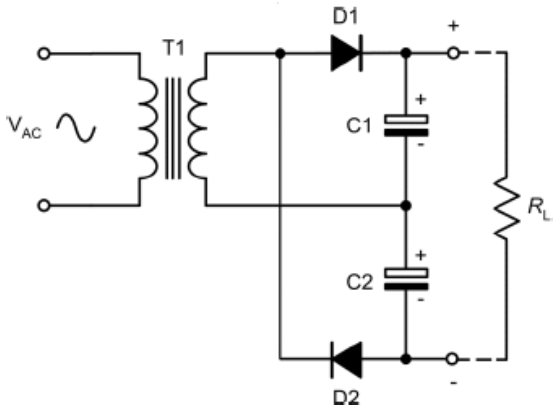


Figure 1.19 A voltage doubler

- In this arrangement C1 will charge to the positive peak secondary voltage while C2 will charge to the negative peak secondary voltage.
- Since the output is taken from C1 and C2 connected in series the resulting output voltage is twice that produced by one diode alone.
- The voltage doubler can be extended to produce higher voltages using the cascade arrangement shown in Fig. 1.20.

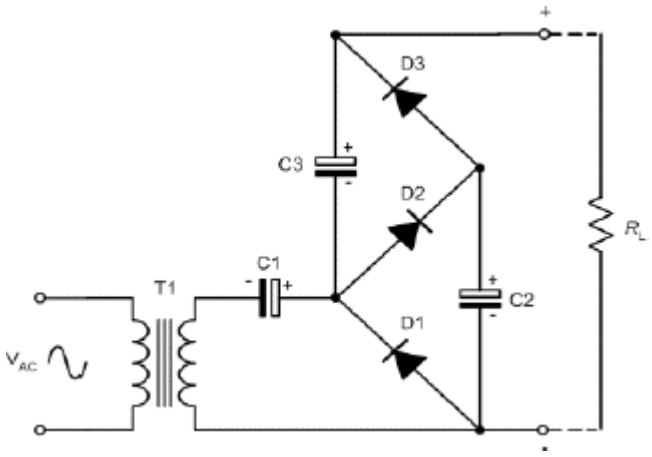


Figure 1.20 A voltage Tripler

- Here C1 charges to the positive peak secondary voltage, while C2 and C3 charge to twice the positive peak secondary voltage.
- The result is that the output voltage is the sum of the voltages across C1 and C3 which is three times the voltage that would be produced by a single diode.
- The ladder arrangement shown in Fig. 1.18 can be easily extended to provide even higher voltages but the efficiency of the circuit becomes increasingly impaired and high-order voltage multipliers of this type are only suitable for providing relatively small currents.

Numerical:

1. An AC supply of 230V is applied to HWR through a transformer of turns ratio 10:1. Find a) dc output voltage and b) Peak Inverse Voltage (PIV). Assume ideal diode

Soln.:

The given 230V is rms voltage at primary denoted by $V_{p(rms)}$ then Maximum primary voltage,

$$V_{p(m)} = \sqrt{2} \times V_{p(rms)} = \sqrt{2} \times 230 = 325.26V$$

Maximum secondary voltage $V_{s(m)}$ is calculated as

$$\frac{V_{p(m)}}{V_{s(m)}} = \frac{N_p}{N_s} \Rightarrow V_{s(m)} = V_{p(m)} \times \frac{N_s}{N_p} \quad \therefore V_{s(m)} = 325.26 \times \frac{1}{10} = \mathbf{32.526V}$$

1. DC output voltage V_{dc}=0.3183$V_{s(m)}$= 0.3183 X 32.526=10.353V
2. PIV of HWR = the maximum value of the secondary voltage= $V_{s(m)}$=32.526V

2. In a HWR the input voltage is 230V and transformer ratio is 2:1. Determine the maximum and Average values of power delivered to the load. Assume R_L=200Ω

Soln.:

The given 230V is rms voltage at primary denoted by $V_{p(rms)}$ then Maximum primary voltage,

$$V_{p(m)} = \sqrt{2} \times V_{p(rms)} = \sqrt{2} \times 230 = 325.26V$$

∴Maximum secondary voltage $V_{s(m)}$is

$$\frac{V_{p(m)}}{V_{s(m)}} = \frac{N_p}{N_s} \Rightarrow V_{s(m)} = V_{p(m)} \times \frac{N_s}{N_p} \quad \therefore V_{s(m)} = 325.26 \times \frac{1}{2} = \mathbf{162.63V}$$

2. Average value of power delivered to the load

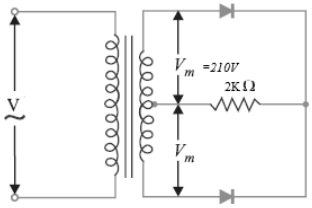
$$P_{dc} = \frac{(0.3183\, V_{s(m)})^2}{R_L} = \frac{(0.3183 \times 162.63)^2}{200} = 13.398W$$

3. A FWR has a load of 1kΩ. The AC voltage applied is 200-0-200V. Calculate a) dc voltage b) dc current c) ripple voltage. Assume ideal diodes. Draw the circuit

Soln:

4. In a c-t FWR, R_L is 2kΩ. Each diode has forward resistance of 10Ω. The peak value of voltage across secondary is 210V. Find the peak value of current, rms value of current, and ripple factor. Draw the circuit.

Soln: Given: $v_m = 210V, r_f = 10\Omega, R_L = 2k\Omega$



At any instant in a C-T FWR one diode is conducting

∴ Circuit resistance = $r_f + R_L$

1. $i_m = \frac{v_m}{r_F + R_L} = \frac{210}{10+2\times10^3} = 0.1044A$

2. $i_{rms} = \frac{i_m}{\sqrt{2}} = \frac{0.1044}{\sqrt{2}} = 0.073A$

3. $\gamma = \sqrt{\left(\frac{i_m/\sqrt{2}}{2i_m/\pi}\right)^2 - 1} = \sqrt{\left(\frac{0.1044/\sqrt{2}}{2\times0.1044/\pi}\right)^2 - 1} = \sqrt{\left(\frac{0.0738}{0.0664}\right)^2 - 1} = \sqrt{0.2353} = 0.48$

5. For a Zener Regulator, if Vz=10V, Rs=1kΩ, RL=2kΩ, and the input voltage varies from 22V to 40V. Find the maximum and minimum values of Zener current. Draw the circuit.

Soln:

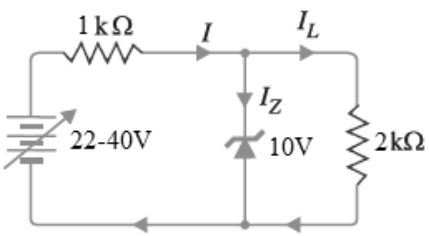


Fig-1

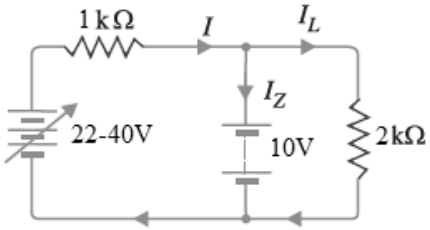


Fig-2

The first step is to determine the state of the zener diode. It is easy to see that for the given range of voltages (22-40V), the voltage across the zener is greater than V_Z (= 10 V). Hence the zener diode will be in the "on" state for this range of applied voltages. Consequently, it can be replaced by a battery of 10 V as shown in Fig-2.

Maximum zener current: The zener will conduct maximum current when the input voltage is maximum i.e. 40 V. Under such conditions:

Voltage across 1KΩ=40-10=30V

Current through 1KΩ, $I = \frac{30}{1k\Omega} = 30mA$

Load current, $I_L = \frac{10}{2k\Omega} = 5mA$

By KCL, $I = I_L + I_Z$, Maximum Zener Current $I_{Z(max)} = I - I_L = 30-5 = 25mA$

Minimum Zener current: The zener will conduct minimum current when the input voltage is minimum i.e. 22 V. Under such conditions, we have,

Voltage across 1KΩ=22-10=12V

Current through 1KΩ, $I = \frac{12}{1k\Omega} = 12mA$

Load current, $I_L = 5mA$

By KCL, $I = I_L + I_Z$, Minimum Zener Current $I_{z(min)} = I - I_L = 12 - 5 = 7mA$

2.1. Common-Emitter (CE) amplifier:

a. Without feedback:

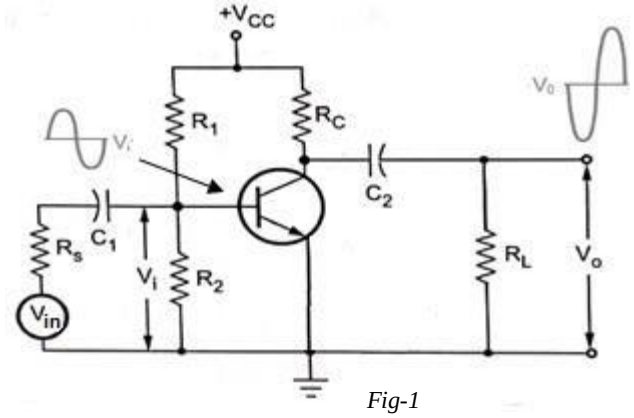


Fig-1

- Fig-1shows a simple common emitter circuit which uses an NPN transistor whose:
  - Collector terminal (output terminal) is connected to supply voltage V_{CC} through the collector resistor R_C.
  - Base terminal is provided with the AC signal which needs to be amplified.
  - Emitter terminal is grounded (hence also referred to as Grounded Emitter configuration).
- In this arrangement, as the input voltage V_{in} increases, the base current I_B also increases which in turn increases the collector current I_C.This causes an increase in the voltage drop across the collector resistor R_C which results in a decreased output voltage V_0 as given by $V_0 = V_{cc} - I_C R_C$
- Similarly as the input voltage goes on decreasing, I_B and hence I_C decrease, due to which the voltage drop across R_C also decreases thereby increasing the output voltage. This indicates that for the positive half-cycle of the input waveform, amplified negative half-cycle is obtained and for the negative input pulse, the output will be an amplified positive pulse. Hence there exists a phase-shift of 180° between the input and the output waveforms of the common emitter amplifier for which it is also referred to as Inverting Amplifier.
- This common-emitter amplifier, inverts the input signal as it is amplified. In other words, a positive-going input voltage causes the output voltage to decrease, or move toward the negative, and vice versa.
- In other words, with no feedback resistor, V_{BE} equals input (V_{in}). Therefore, for instance if input V_{in}increases by 100mV, then V_{BE}increases by 100mV: a change in one is the same as a change in the other since the two voltages are equal to each other.

b. With feedback.

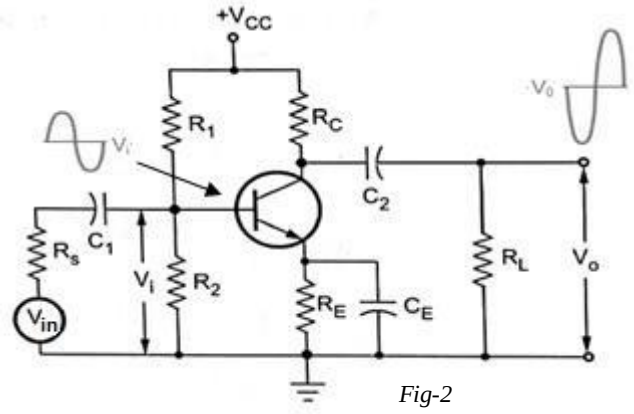


Fig-2

- The amplifier configuration shown in fig-2 is a common-emitter with feedback resistor R_E connected between emitter and ground.
- The new feedback resistor (R_E) drops voltage proportional to the emitter current through the transistor, and it does this to oppose the input signal influence on the base-emitter junction of the transistor.
- With $R_E$in the V_{in}- V_{BE} loop, V_{BE} will no longer be equal to V_{in}. The $R_E$will drop a voltage proportional to emitter current, which is in turn controlled by the base current, which is in turn controlled by the voltage dropped across the base-emitter junction of the transistor (V_{BE}). Thus, if V_{in} increases in a positive direction, it would increase V_{BE}, causing more base current, causing more collector (load) current, causing more emitter current, and causing more feedback voltage to be dropped across R_E. This increase of voltage drop across the feedback resistor,

though, *subtracts* from V_{in} to reduce the V_{BE}, so that the actual voltage increase for V_{BE} will be less than the voltage increase of V_{in}.

- The emitter bypass capacitor C_E when added into the circuit, increases its gain considerably by short-circuiting the emitter resistance R_E for high frequency signals, which results in the reduction of the overall transistor load.
- No longer will a 100mV increase in V_{in} result in a full 100mV increase for V_{BE}, because the two voltages are *not* equal to each other.
- Consequently, the input voltage has less control over the transistor than before, and the voltage gain for the amplifier is reduced due negative feedback.
- In practical common-emitter circuits, negative feedback is a necessity for stable operation. A common-emitter transistor amplifier with no negative feedback, has the full amplitude of V_{in} impressed across the transistor's base-emitter-junction. This gives a large voltage gain. Unfortunately, the relationship between base-emitter voltage and base-emitter current changes with temperature, as predicted by the "diode equation". As the transistor heats up, there will be less of a forward voltage drop across the base-emitter junction for any given current. This causes a problem, as the R1/R2 voltage divider network is designed to provide the correct quiescent current through the base of the transistor so that it will operate in desired class of operation. If the transistor's voltage/current relationship changes with temperature, the amount of DC bias voltage necessary for the desired class of operation will change. A hot transistor will draw more bias current for the same amount of bias voltage, making it heat up even more, drawing even more bias current. The result, if unchecked, is called ***thermal runaway*****.**

Need of Negative Feedback in Amplifiers

1. Stabilizes Amplifier Gain

- The negative feedback stabilizes the gain of the amplifier by reducing the dependence of amplifier gain on various transistor parameters or variation in the supply voltage.

2. Reduces Non-linear Distortion

- The use of negative feedback also reduces the non-linear distortion level in the large signal amplifiers.

3. Increases Circuit Stability

- The output of an amplifier without negative feedback is affected by the variations in the temperature, frequency, or amplitude of the signal which further changes the gain of the amplifier and as result, a distorted signal is obtained at the output. Hence, the negative feedback is applied so that the gain of the amplifier is stabilized.

4. Increases Input Impedance/Resistance

- The use of negative feedback increases the input impedance or resistance of the amplifier.

5. Decreases Output Impedance/Resistance

- The use of negative feedback decreases the output impedance or resistance of the amplifier.

6. Reduces Noise Level

- The negative feedback applied to the amplifiers is in the opposite phase to that of the applied input signal, hence it cancels out the noises which are introduced in the output

signal by the amplifier circuit. As a result, one gets the output signal with a reduced noise level.

7. Improves Frequency Response & Bandwidth

- The negative feedback which is applied to the amplifiers is a resistive network, hence the gain of the amplifier with negative feedback is independent of signal frequency. As a result, the gain becomes constant over a wide range of signal frequencies in this way the frequency response of the amplifier with negative feedback is improved.

8. More Linear Operations

- In the case of an ordinary amplifier, one gets a very high value of the output signal even for a small value of the input signal hence the output signal is not proportional to the applied input signal. But in the case of an amplifier with negative feedback, every parameter is controlled by the applied feedback network. Hence in the case of an amplifier with negative feedback, the relation between the output and the input signal is more linear.

Note: The main disadvantage of using negative feedback in amplifiers is that, the overall gain of the amplifier is reduced by the factor of $(1+\beta A_v)$.

2.2. Multi-stage amplifiers:

- In order to provide sufficiently large values of gain, it is frequently necessary to use a number of interconnected stages within an amplifier. The overall gain of an amplifier with several stages(i.e. a multi-stage amplifier) is simply the product of the individual voltage gains. Hence: $A_V = A_{V1} \times A_{V2} \times A_{V3}$
- Note, however, that the bandwidth of a multistage amplifier will be less than the bandwidth of each individual stage. In other words, an increase in gain can only be achieved at the expense of a reduction in bandwidth.
- Signals can be coupled between the individual stages of a multi-stage amplifier using one of a number of different methods shown in Fig-3.

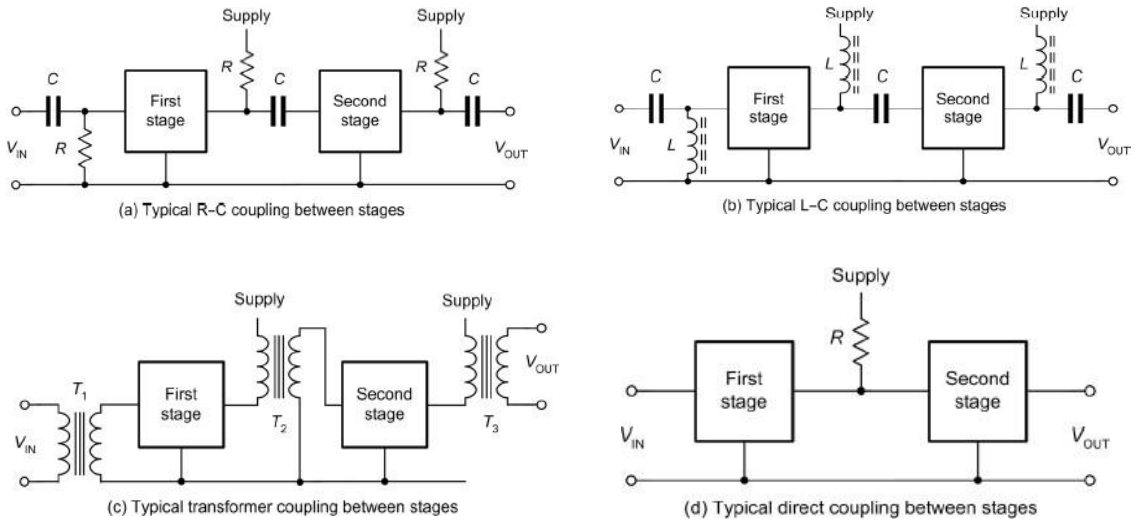


Fig-3

- The most commonly used method is that of R–C coupling as shown in In Fig-3(a). In this coupling method, the stages are coupled together using capacitors having a low reactance at the signal frequency and resistors (which also provide a means of connecting the supply). Fig-4 shows a practical example of this coupling method.

- A similar coupling method, known as L–C coupling, is shown in Fig-3(b). In this method, the inductors have a high reactance at the signal frequency. This type of coupling is generally only used in RF and high-frequency amplifiers.
- Two further methods, transformer coupling and direct coupling, are shown in Fig-3(c) and 3(d), respectively. The latter method is used where d.c. levels present on signals must be preserved.

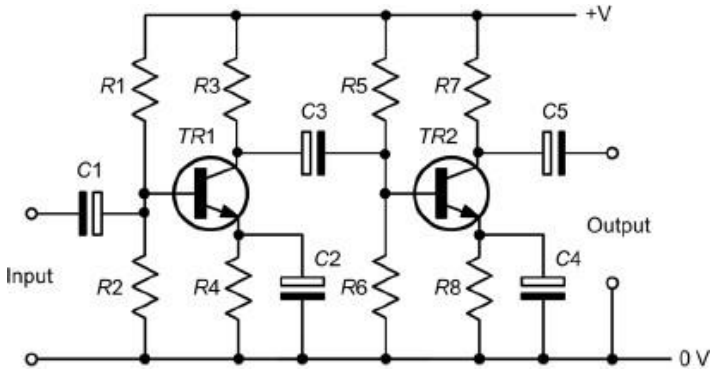


Fig-4: A typical two-stage high-gain R–C coupled common-emitter amplifier

2.3. Transistor as a Switch:

- Transistor as a switch operates in two regions i.e., saturation region (fully-ON) and the cutoff region (fully-OFF).

Cut-off Region

- The operating conditions of the transistor are zero input base current (I_B=0), zero output collector current(I_C=0), and maximum collector voltage (V_{CE}) which results in a large depletion layer and no current flowing through the device as in below fig-5

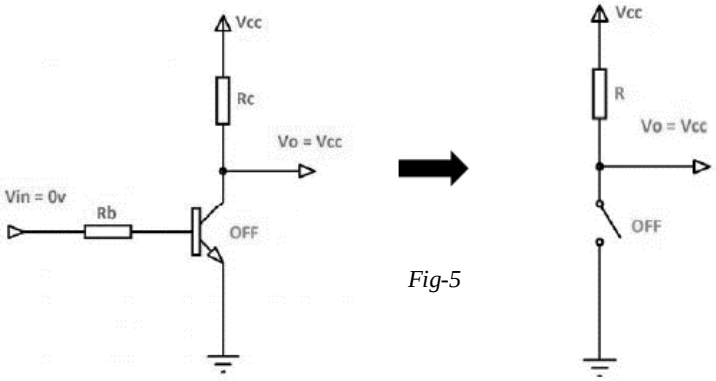


Fig-5

Saturation Region

- In this region, the transistor will be biased so that the maximum amount of base current (I_B) is applied, resulting in maximum collector current(I_C=V_{CC}/R_L) and then resulting in the minimum collector-emitter voltage (VCE ~ 0) drop. At this condition, the depletion layer becomes as small as the possible and maximum current flows through the transistor. Therefore the transistor is switched "Fully-ON" as in fig-6

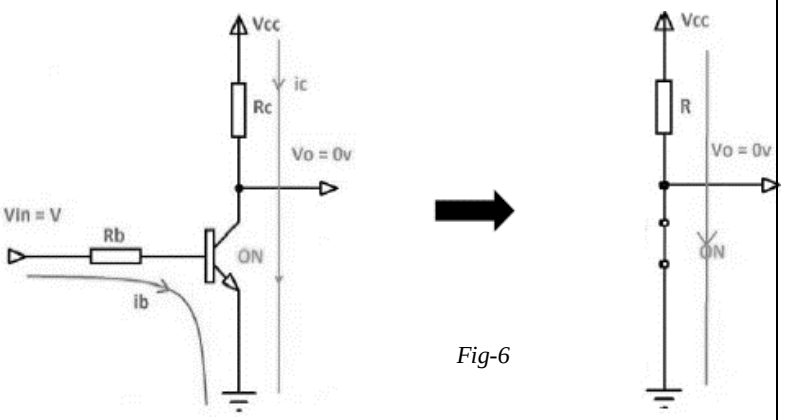


Fig-6

Module-2:

Oscillators and Operational Amplifiers

Oscillators

- An oscillator is a circuit which produces a continuous, repeated, alternating waveform without any input.
- Oscillators basically convert unidirectional current flow from a DC source into an alternating waveform which is of the desired frequency, as decided by its circuit components.
- Practically, the oscillators are nothing but the amplifier circuits which are provided with a positive or regenerative feedback wherein a part of the output signal is fed back to the input (Figure 3). Here the amplifier consists of an amplifying active element which can be a transistor or an Op-Amp and the back-fed in-phase signal is held responsible to keep-up (sustain) the oscillations by making-up for the losses in the circuit.

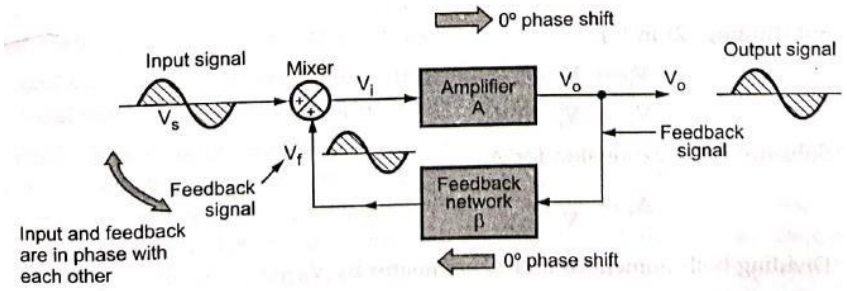


Fig-1: Typical oscillator

- In above figure, amplifier gain is A. It amplifies Vi, A times to produce Vo

$$\therefore A = \frac{V_O}{V_i}$$

This is called open loop gain of the amplifier

- For overall circuit, the ratio of output V_o to input V_s considering effect of feedback is called closed loop gain Af

$$\therefore A_f = \frac{V_O}{V_s}$$

- The feedback is positive and voltage V_f is added to V_s to generate input of amplifier V_i, then

$$V_i = V_s + V_f$$

- The feedback voltage depends on the feedback element gain β, then

$$V_f = \beta V_o$$

- Substituting value of V_f in V_i equation, then

$$V_i = V_s + \beta V_o,$$

$$V_s = V_i - \beta V_o$$

- Substituting expression of A_f ,

$$\therefore A_f = \frac{V_O}{\cancel{V_i - \beta V_o}}$$

- Dividing numerator and denominator by V_i,V_O/V_i

$$\therefore A_f = \frac{\cancel{V_i/V_i}}{\cancel{V_i/V_i - \beta V_o/V_i}}$$

$$\therefore \boldsymbol{A_f = \frac{A}{1 - A\beta}}$$

- Considering the various values of β and corresponding values of A_f, for constant amplifier gain of A=20

A	β	A_f
20	0.005	22.22
20	0.04	100
20	0.045	200
20	0.05	∞

- From the values in above table, gain with feedback increases as the amount of positive feedback increases and becomes infinite. This indicates that circuit can produce output without external input (V_s=0) just by feeding part of the output as its own input. But output cannot be infinite instead gets driven into the oscillations. In general, circuit stops amplifying and starts oscillating.

- If Aβ > 1, then the oscillations will increase in amplitude (Fig-2); while if Aβ < 1, then the oscillations will be damped (Fig-3). On the other hand, Aβ = 1 leads to the oscillations which are of constant amplitude (Fig-4).

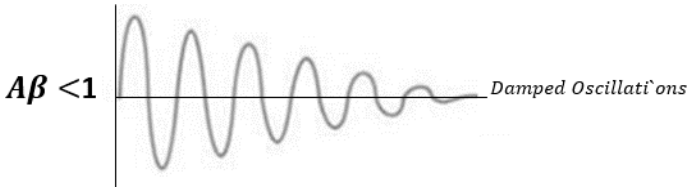


Fig-2

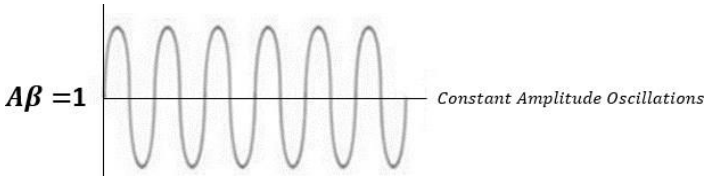


Fig-3

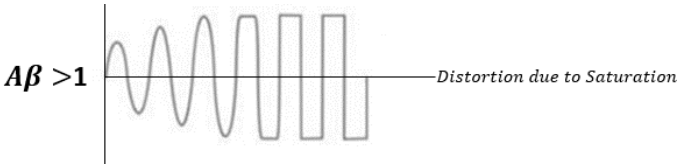


Fig-4

- For constant amplitude (sustained) oscillation the necessary and sufficient conditions are
  1. ***The feedback must be positive (i.e. the signal fed back must arrive back in-phase with the signal at the input); 0° or 360°***
  2. ***The overall loop voltage gain must be greater than 1 (i.e. the amplifier's gain must be sufficient to overcome the losses associated with any frequency selective feedback network).Aβ=1***

The above two conditions are called ***Barkhausen criterion.***

1. Ladder network oscillator:

- RC phase shift oscillator or simply RC oscillator is a type of oscillator where a simple RC network (resistor-capacitor) network is used for giving the required phase shift to the feedback signal.
- The circuit diagram of a simple single stage RC network is shown in the fig-5.



Fig-5: Single stage RC phase shift network with input & output waveforms

- Theoretically in a simple RC circuit, the output voltage will lead the input voltage by a phase angle Φ =90°. But in practical, the phase angle will be below 90° as it is impossible to get a purely ideal capacitor. Phase shift of a practical RC network depends on the value of the capacitor, resistor and the operating frequency.
- Fig-5 also shows the input & output signals differing in phase. A simple phase-shift oscillator based on a three stage C–R ladder network using op-amp is shown in Fig-1
- R-C phase shift oscillator uses an op-amp in inverting amplifier mode. Thus it introduces the phase shift of 180° between input and output. The feedback network consists of 3 RC sections each producing 60° phase shift. Such an RC phase shift oscillator using op-amp is shown in Fig-6.

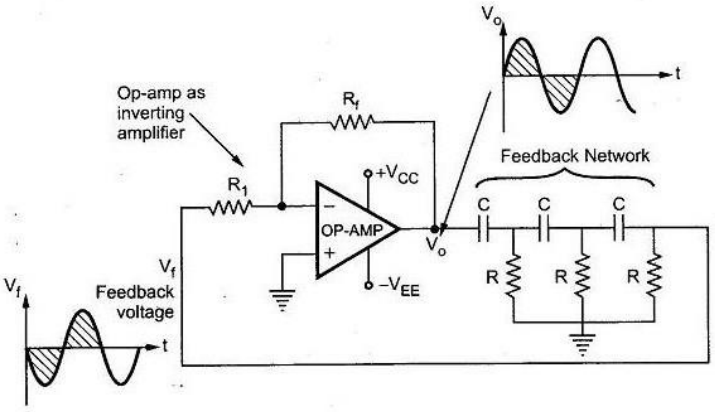


Fig-6: R-C phase shift oscillator using op-amp

- The output of the amplifier is given to the feedback network. The output of the feedback network drives the amplifier. The total phase shift around a loop is 180° of the amplifier and 180° due to 3 RC sections, thus 360°. This satisfies the required condition for positive feedback and circuit works as an oscillator.
- The generalized expression for the frequency of oscillations produced by a RC phaseshift oscillator is given by

$$f_r = \frac{1}{2\pi RC\sqrt{2N}}$$

Where, N is the number of RC stages formed by the resistors R and the capacitors C

- Here N=3 then

$$\boldsymbol{f_r = \frac{1}{2\pi RC\sqrt{6}}}$$

- At this frequency, β calculates to be equal to –(1/29), where the negative sign indicates a phase shift of 180°

$$\beta = |\frac{1}{29}|$$

- Now to have the oscillations,

$$|A\beta| \geq 1$$

$$|A||\beta| \geq 1$$

$$|A| \geq \frac{1}{|\beta|} \geq 29$$

- Thus circuit will work as an oscillator which will produce a sinusoidal waveform if the gain is 29 and total phase shift around a loop is 360°. This satisfies the Barkhausen criterion for the oscillator. These oscillators are used over the audio frequency range i.e. about 20 Hz up to 100 kHz.

2. Wien-Bridge Oscillator:

- A Wien-Bridge Oscillator is a type of phase-shift oscillator which is based upon a Wien-Bridge network (Fig-7) comprising of four arms connected in a bridge fashion.
- Here two arms are purely resistive while the other two arms are a combination of resistors and capacitors.In particular, one arm has resistor and capacitor connected in series (R1 and C1) while the other has them in parallel (R2 and C2). This indicates that these two arms of the network behave identical to that of high pass filter or low pass filter, mimicking the behaviour of the circuit shown by Fig-8.

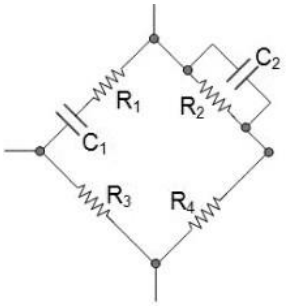


Fig-7: Wein-bridge network

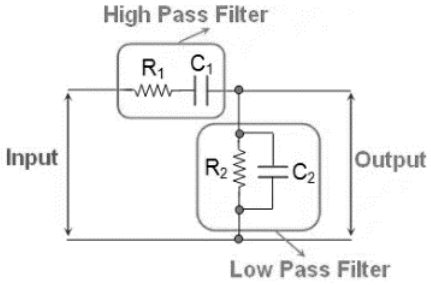


Fig-8: Two arms of Wein-bridge network

- In this circuit, at high frequencies, the reactance of the capacitors C1 and C2 will be much less due to which the voltage V_0 will become zero as R2 will be shorted.
- Next, at low frequencies, the reactance of the capacitors C1 and C2 will become very high. However even in this case, the output voltage V0 will remain at zero only, as the capacitor C1 would be acting as an open circuit.
- This kind of behaviour exhibited by the Wien-Bridge network makes it a lead-lag circuit in the case of low and high frequencies, respectively.
- Amidst two high and low frequencies, there exists a particular frequency at which the values of the resistance and the capacitive reactance will become equal to each other, producing the maximum output voltage.
- This frequency is referred to as resonant frequency. The resonant frequency for a Wein Bridge Oscillator is calculated using the following formula:

$$f_r = \frac{1}{2\pi\sqrt{R_1C_1R_2C_2}}$$

If R_1=R_2=R and C_1=C_2=C then

$$f_r = \frac{1}{2\pi RC}$$

- Further, at this frequency, the phase-shift between the input and the output will become zero and the magnitude of the output voltage will become equal to one-third of the input value. In addition, it is seen that the Wien-Bridge will be balanced only at this particular frequency.
- Wien-bridge oscillators designed using Op-Amps as a part of their amplifier section, is as shown by Fig-9.

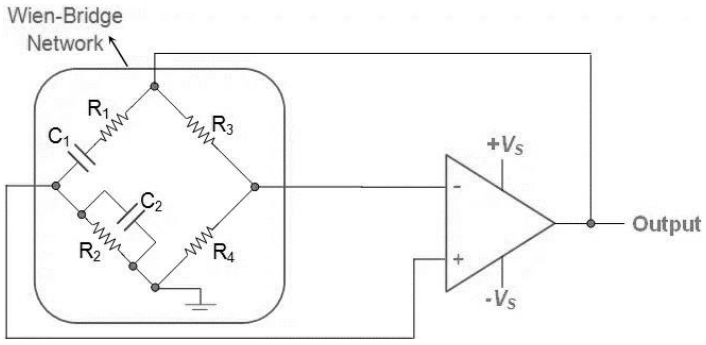


Fig-9: Wein-Bridge Oscillator using Op-amp

- However it is to be noted that, here, the Op-Amp is required to act as a non-inverting amplifier as the Wien-Bridge network offers zero phase-shift. Further, from the circuit, it is evident that the output voltage is fed back to both inverting and non-inverting input terminals.
- At resonant frequency, the voltages applied to the inverting and non-inverting terminals will be equal and in-phase with each other.
- However, even here, the voltage gain of the amplifier needs to be greater than 3 to start oscillations and equal to 3 to sustain them. In general, these kind of Op-Amp-based Wien Bridge Oscillators cannot operate above 1 MHz due to the limitations imposed on them by their open-loop gain.
- Wien-Bridge networks are low frequency oscillators which are used to generate audio and sub-audio frequencies ranging between 20 Hz to 20 KHz.
- Further, they provide stabilized, low distorted sinusoidal output over a wide range of frequency which can be selected using decade resistance boxes.
- In addition, the oscillation frequency in this kind of circuit can be varied quite easily as it just needs variation of the capacitors C1 and C2.

3. Multivibrators

- Multivibrators are a family of oscillator circuits that produce output waveforms consisting of one or more rectangular pulses. The term ‘multivibrator’ simply originates from the fact that this type of waveform is rich in harmonics (i.e. ‘multiple vibrations’).
- Multivibrators use regenerative (i.e. positive) feedback; the active devices present within the oscillator circuit being operated as switches, being alternately cut-off and driven into saturation. The principal types of multivibrator are:
  a. Astable multivibrators that provide a continuous train of pulses (these are sometimes also referred to as free-running multivibrators);
  b. Monostable multivibrators that produce a single output pulse (they have one stable state and are thus sometimes also referred to as‘one-shot’);
  c. Bistable multivibrators that have two stable states and require a trigger pulse or control signal to change from one state to another.

3.a. Astable Multivibrators:

- Initially, assuming that the output of the circuit is in positive saturation voltage because of resistor R as feedback, the current will start flowing through the resistor R, and the capacitor will start charging slowly.

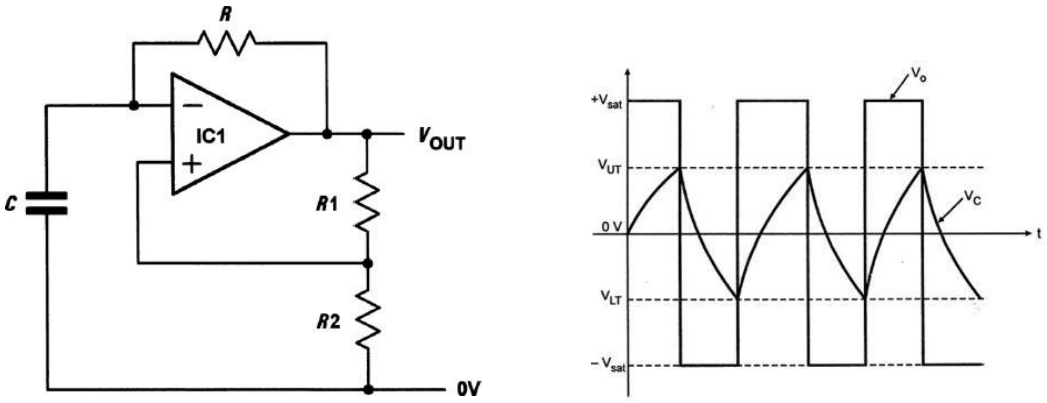


Fig-10: Astable Multivibrators using op-amp

- As can been seen in the fig-10, it is shown with the black dotted line. When the capacitor charge reaches the upper threshold voltage, the output will switch from positive saturation voltage to negative saturation voltage. When that happens, the capacitor will start discharging towards the negative saturation voltage.
- Now when the voltage at the non-inverting terminal is slightly more than the inverting terminal, the output will again switch from negative saturation voltage to positive saturation voltage. This way by the charging and discharging process, this circuit can generate the Astable signal at the output.
- In this circuit, the time-period is dependent upon the value of the resistor and capacitor. It also depends upon the upper and lower threshold voltage of the op-amp. This is how an Op-amp based Astable multivibrator circuit works.
- The upper threshold voltage (i.e. the maximum positive value for the voltage at the inverting input) will be given by:

$$V_{UT} = V_{CC} \times \frac{R_2}{R_1 + R_2}$$

- The lower threshold voltage (i.e. the maximum negative value for the voltage at the inverting input) will be given by:

$$V_{LT} = -V_{CC} \times \frac{R_2}{R_1 + R_2}$$

- Finally, the time for one complete cycle of the output waveform produced by the astable oscillator is given by:

$$\boldsymbol{T = 2CR\, ln\,(1 + 2\,(\frac{R_2}{R1}))}$$

Op-amp Characteristics:

Characteristics	Ideal Op-Amp	Practical Op-Amp
1. Infinite Voltage Gain	• An ideal op amp will have ***infinite voltage gain***. • An ideal op amp will produce mega-gain, practically, produce infinite gain. • It will amplify the signal infinite times which is never needed.	• Apractical op-amp can only produce a finite gain.
2. Infinite Input impedance	• An ideal op amp will have ***infinitely high input impedance***. • The higher the impedance, the lower the current that an op-amp draws. • A high input impedance is required so that the op amp does not disturb the original circuit by pulling current from it.	• A practical op amp has finite input impedance.
3. Zero Output Impedance	• An ideal op amp will have ***zero output impedance***. • When an op amp produces its output signal, op amp should have a zero voltage so that the maximum voltage will be transferred to the output load.	• A practical op amp will always have some output impedance, though it is low. A typical value can be 75Ω.
4. Gain Independent of Frequency	• In an ideal op amp, the ***gain*** that the op amp produces will be ***independent of frequency***. • This means that regardless of the frequency of the input signal going into the op amp, the gain that is produced will be constant and good across all frequencies.	• In practical op amps, the gain that is produced is only for a certain bandwidth of frequencies. Outside of this bandwidth, the gain that the op amp produces will decline.
5. Zero Input Voltage Offset	• In an ideal op amp, if ***no voltag***e is applied to the inverting and noninverting input pins, the ***op amp will output a voltage of zero (0***), since there is no difference at all of the voltage applied to the two input pins.	• A practical op amp will have slight offset even if the voltage applied to the pins are the same. To correct this offset, voltage must be applied to the offset pin.
6. Output Swings to positive and negative voltage supply rails	• In an ideal op amp, the ac voltage which is fed into the op amp to be amplified will swing all the way up for the DC positive supply rail and all the way down for the DC negative supply rail, making 100% efficient use of the DC voltage supplied to an op amp.	• In practical op amps, the amplified signal will not fully reach the DC supply rails. They will fall short of it.
7. Output swings instantly to the correct value	• In an ideal op amp, the output will swing instantly to the amplified voltage value. There will be no time delay between the time the voltage is input into the op amp till the time it is output. It will all be instantaneous.	• In practical op amps, the amplified signal will take time to reach the fully amplified voltage value. This is determined by the slew rate of the op amp.

- An operational amplifier, or op-amp, is a very high gain differential amplifier with high input impedance and low output impedance. Typical uses of the operational amplifier are to provide voltage amplitude changes (amplitude and polarity), oscillators, filter circuits, and many types of instrumentation circuits. An op-amp contains a number of differential amplifier stages to achieve a very high voltage gain.

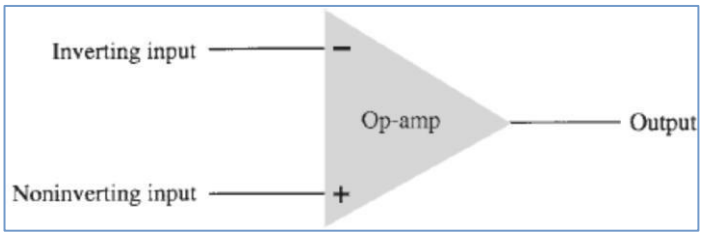
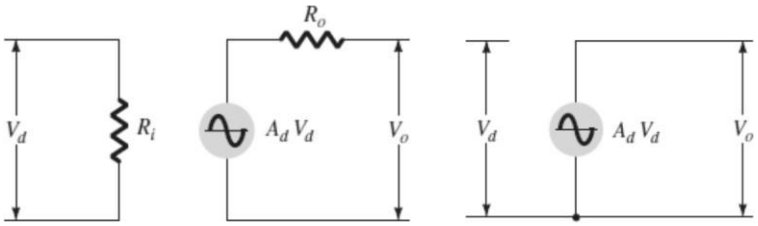


Fig-1: Basic op-amp.



(a) Practical *(b) ideal*

Fig-2: AC equivalent of op-amp circuit:

- Fig-1 shows a basic op-amp with two inputs and one output as would result using a differential amplifier input stage. Each input results in either the same or an opposite polarity (or phase) output, depending on whether the signal is applied to the plus (+) or the minus (-) input, respectively
- The ac equivalent circuit of the op-amp is shown in Fig-2. As shown, the input signal applied between input terminals sees an input impedance Ri that is typically very high. The output voltage is shown to be the amplifier gain times the input signal taken through an output impedance Ro, which is typically very low. An ideal op-amp circuit, as shown in Fig-2b, would have infinite input impedance, zero output impedance, and infinite voltage gain.

Other parameters related to op-amp

- **Differential gain 'Ad':**
  The gain with which differential amplifier amplifies the difference between two input signals

$$A_D = \frac{V_O}{V_d} = \frac{V_O}{V_1 - V_2}$$

- **Common Mode Gain 'Ac':**
  For an ideal op-amp the output of the differential amplifier is zero, however for practical op-amp the output of the differential amplifier is not only depends on difference of two input signal but also depends on average of two input signals

$$A_C = \frac{V_O}{V_C} = \frac{V_O}{(V_1 + V_2)/2}$$

- **Common Mode Rejection Ratio 'ρ':**

$$\rho = \frac{A_d}{A_C}$$

- **Slew rate:** Slew rate of an op-amp is defined as the maximum rate of change of the output voltage due to a step input voltage.

Virtual ground in op-amp:

- In op-amp the term virtual ground means that the voltage at a particular node is almost equal to ground voltage (0V), this is because of feedback due to R_2 and the high gain of the op amp as in fig-3.It is not physically connected to ground. This concept is very useful in analysis of op-amp circuits and it will make a lot of calculations very simple.

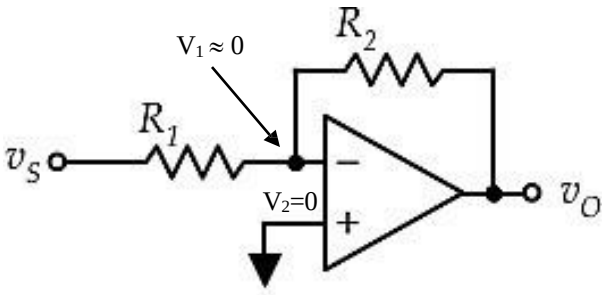


Fig-3

- Since Zin = ∞, current I flowing through op-amp is zero and voltage V_1at inverting pin / node is equal to voltage V_2(=0) at non-inverting pin, by virtual ground concept.

2. Inverting Amplifier:

- An inverting amplifier (also known as an inverting operational amplifier or an inverting op-amp) is a type of operational amplifier circuit which produces an output which is out of phase with respect to its input by 180º. This means that if the input pulse is positive, then the output pulse will be negative and vice versa.
- The input signal is applied to the inverting terminal of the op-amp via the resistor Ri. Non-inverting terminal is grounded. The necessary feedback to stabilize the circuit and to control the output, is provided through a feedback resistor $R_f$as in fig-4.
- By virtual ground concept, $V_1 = V_2 = 0$
- Mathematically the voltage gain offered by the circuit is given as

$$A_V = \frac{V_{out}}{V_{in}}$$

Currents I_i& I_f are given by,

$$I_i = \frac{V_i - V_1}{R_i} = \frac{V_i - 0}{R_i} = \frac{V_i}{R_i}$$

and

$$I_f = \frac{V_1 - V_o}{R_f} = \frac{0 - V_o}{R_f} = -\frac{V_o}{R_f}$$

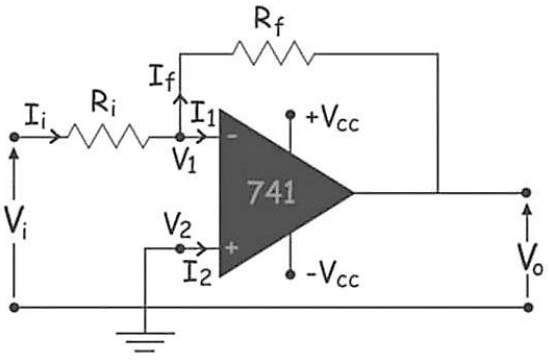


Fig-4

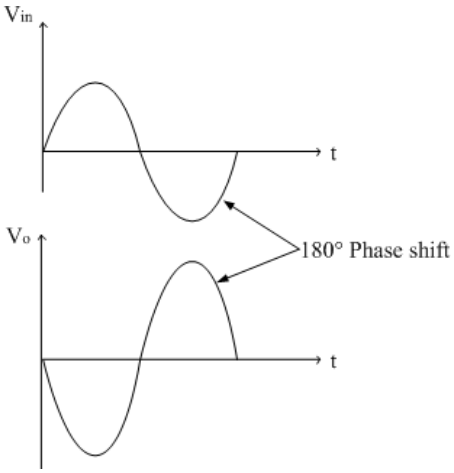
- Ideal op amp has infinite input impedance due to which the currents flowing into its input terminals are zero i.e. $I_1 = I_2 = 0$, Thus, $I_i = I_f$.

$$\frac{V_i}{R_i} = -\frac{V_o}{R_f}$$

$$\frac{V_o}{V_i} = -\frac{R_f}{R_i}$$

$$V_o = -\frac{R_f}{R_i} V_i$$

$$V_o = A_V V_i$$



$$\therefore \boldsymbol{A_V = -\frac{R_f}{R_i}}$$

- This indicates that the voltage gain of the inverting amplifier is decided by the ratio of the feedback resistor to the input resistor with the minus sign indicating the phase-reversal.

3. Non-Inverting Amplifier:

- Non inverting amplifier is an op amp based amplifier with positive voltage gain.
- When any signal is applied to the non – inverting input, it does not change its polarity when it gets amplified at the output terminal. So, in that case, the gain of the amplifier is always positive.
- As in fig-5, By virtual ground concept,
  $V_1 = V_2 = V_i$
- Mathematically the voltage gain offered by the circuit is given as

$$A_V = \frac{V_{out}}{V_{in}}$$

- Currents I_1& I_f are given by,

$$I_1 = \frac{V_1 - 0}{R_i} = \frac{V_i - 0}{R_i} = \frac{V_1}{R_i}$$

and

$$I_f = \frac{V_o - V_1}{R_f} = \frac{V_o - V_i}{R_f}$$

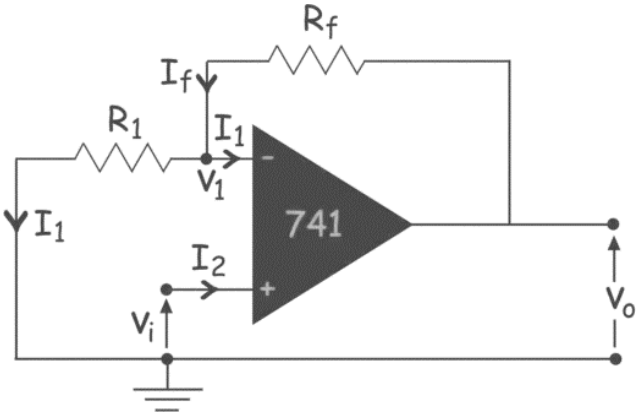
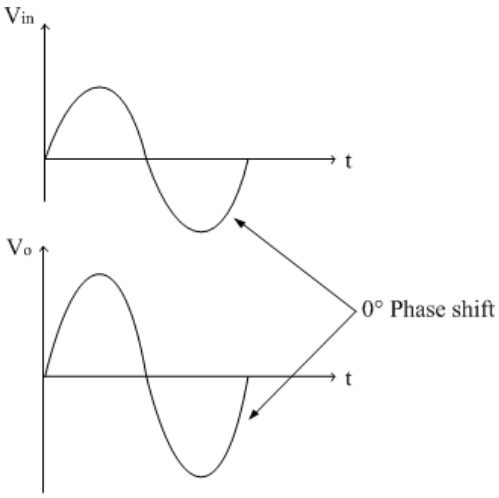


Fig-5



- Ideal op-amp has infinite input impedance due to which the currents flowing into its input terminals are zero i.e. $I_1 = I_2 = 0$, Thus, $I_1 = I_f$.

$$\frac{V_1}{R_i} = \frac{V_o - V_i}{R_f}$$

$$\frac{V_o}{R_f} = \frac{V_i}{R_1} + \frac{V_i}{R_f}$$

$$\frac{V_o}{V_i} = (1 + \frac{R_f}{R_1})$$

$$V_o = (1 + \frac{R_f}{\cancel{)}_1}$$

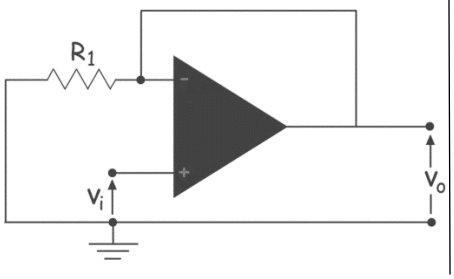
$$V_o = A_V V_i$$

$$\therefore \boldsymbol{A_V = (1 + \frac{R_f}{R_1})}$$

- This term does not contain any negative part. Hence, it proves that the input signal to the circuit gets amplified without changing its polarity at the output.
- From the expression of voltage gain of non-inverting op-amp, it is clear that, the gain will be unity when $R_f = 0$ or $R_1 = \infty$.

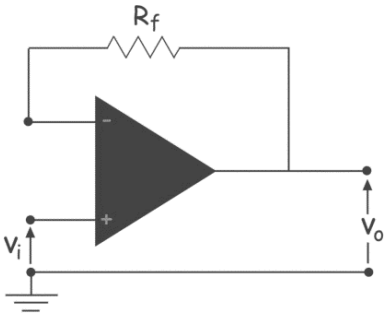
When $R_f = 0$

$$\therefore A_V = (1 + \frac{R_f}{R_1}) = (1 + \frac{0}{R_1}) = 1$$



When $R_1 = \infty$

$$\therefore A_V = (1 + \frac{R_f}{R_1}) = (1 + \frac{R_f}{\infty}) = 1$$



- So, if the feedback path is short circuited and/or open the external resistance of the inverting pin, the gain of the circuit becomes 1.

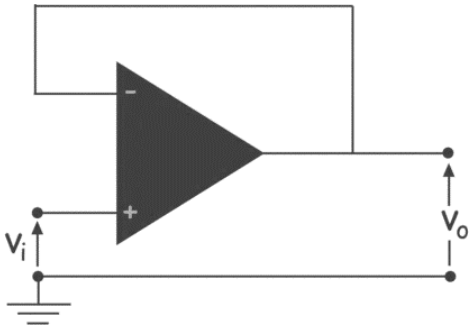
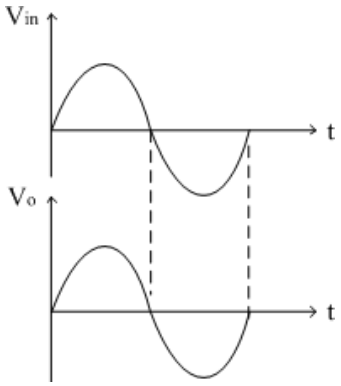


Fig-6: Voltage Follower/Unity Gain Amplifier



- This above circuit is called voltage follower or unity gain amplifier. This is used to isolate two cascaded circuits, because of its infinitely large impedance, at op-amp inputs.

4. Summing Amplifier:

- A summing amplifier is an op-amp circuit that combines several inputs and produces an output that is weighted sum of the inputs.

- Two types:
    1. Inverting:An inverting summing amplifier is one whose output is the inverted sum of the constituent inputs
    2. Non-inverting:A non-inverting summing amplifier is one whose output is the sum of the constituent inputs

3a.Inverting Summing Amplifier:

- Multiple inputs are applied to the inverting input terminal of the Op-Amp, while the non-inverting input terminal is connected to ground. Due to this configuration, the output of voltage Inverting Summing Amplifiercircuit is out of phase by 180° with respect to the input.

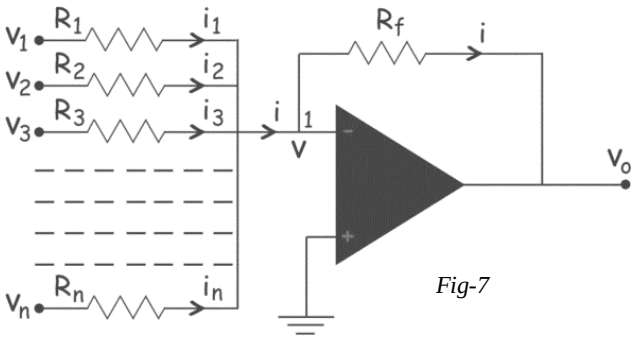


Fig-7

- From the circuit of fig-7, 'i' is the sum of currents of input terminals.

$$i = i_1 + i_2 + i_3 + \ldots\ldots + i_n$$

$$i = \frac{(v_1 - 0)}{R_1} + \frac{(v_2 - 0)}{R_2} + \frac{(v_3 - 0)}{R_3} + \ldots\ldots + \frac{(v_n - 0)}{R_n}$$

$$i = \frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} + \ldots\ldots + \frac{v_n}{R_3}$$

and

$$i_f = \frac{(0 - v_0)}{R_f} = -\frac{v_o}{R_f}$$

- For an ideal op amp, the current at the inverting and non-inverting terminal are zero, and the total current i flows through R_f,

$$\therefore \frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} + \ldots\ldots + \frac{v_n}{R_3} = -\frac{v_o}{R_f}$$

- Output voltage of Inverting Summing Amplifier is given by

$$v_o = -\left(\frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_3 + \cdots + \frac{R_f}{R_n}V_n\right)$$

- This indicates that output voltage v_0 is weighted sum of numbers of input voltages.If $R_1 = R_2 = R_3 = \ldots R_n = R$ then

$$\boldsymbol{v_o = -(V_1 + V_2 + V_3 + \cdots + V_n)}$$

3b.Non-Inverting Summing Amplifier:

- The input voltages are applied to the non-inverting input terminal of the Op-Amp and a part of the output is fed back to the inverting input terminal, through voltage-divider-bias feedback.

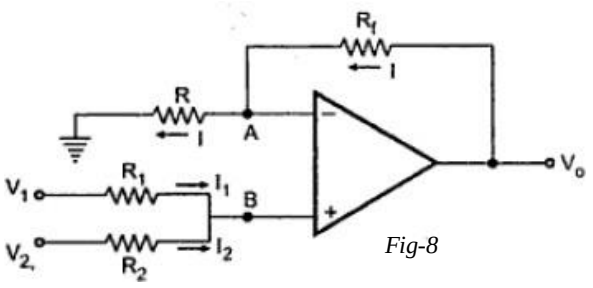


Fig-8

- The circuit of a Non-Inverting Summing Amplifier is shown in fig-8. For the sake of convenience, the following circuit consists of only three inputs, but more inputs can be added.
- Voltage at node A and B are same

$$V_A = V_B$$

- From input side

$$I_1 = \frac{V_1 - V_B}{R_1} \; and \; I_2 = \frac{V_2 - V_B}{R_2}$$

- As input current of op-amp is zero

$$I_1 + I_2 = 0$$

$$\frac{V_1 - V_B}{R_1} + \frac{V_2 - V_B}{R_2} = 0$$

$$\frac{V_1}{R_1} - \frac{V_B}{R_1} + \frac{V_2}{R_2} - \frac{V_B}{2} = 0$$

$$V_B = \frac{R_2 V_1 + R_1 V_2}{R_1 + R_2}$$

- At node A,

$$I = \frac{V_A}{R} = \frac{V_B}{R} \; as \; V_A = V_B \; and \; I = \frac{V_0 - V_A}{R_f} = \frac{V_0 - V_B}{R_f}$$

- Equating above equations

$$\frac{V_B}{R} = \frac{V_0 - V_B}{R_f}$$

$$\frac{V_0}{R_f} = V_B \left[\frac{1}{R} + \frac{1}{R_f}\right]$$

$$V_0 = V_B \left[\frac{R + R_f}{R}\right]$$

- Substituting V_B in above equation,

$$V_0 = \frac{R_2 V_1 + R_1 V_2 \; (R + R_f)}{(R_1 + R_2)}$$

$$V_0 = \frac{R_2 \; (R + R_f)}{(R_1 + R_2)} V_1 + \frac{R_2 \; (R + R_f)}{(R_1 + R_2)} V_2$$

- If $R_1 = R_1 = R = R_f$, then

$$\boldsymbol{V_0 = V_1 + V_2}$$

5. Integrator or Op Amp Integrator:

- An integrator is an op amp circuit, whose output is proportional to the integral of input signal.
- An inverting amplifier can be modified to an **op-amp integrator** circuit, if the feedback resistor R_f is replaced by a capacitor C as in fig-9.
- Applying KCL to node *1*,

$$\frac{V_i}{R} = C \frac{(v_1 - v_o)}{dt}$$

$$\frac{V_i}{R} = C\frac{(0 - v_o)}{dt}$$

$$\frac{V_i}{R} = -C\frac{dv_o}{dt}$$

$$=> dv_o = -\frac{1}{RC}V_i dt$$

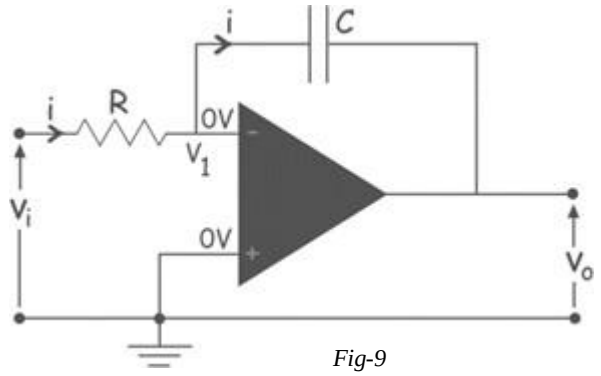


Fig-9

- Integrating both side

$$\boldsymbol{v_o = -\frac{1}{RC}\int V_i\, dt}$$

- Output voltage is the integral function of input voltage.

6. Op-Amp Differentiator:

- Differentiator is an op amp based circuit, whose output signal is proportional to differentiation of input signal.
- An op-amp differentiator is basically an inverting amplifier with a capacitor of suitable value at its input terminal.
- Considering an ideal op amp, applying KCL at node 1 of fig-10,

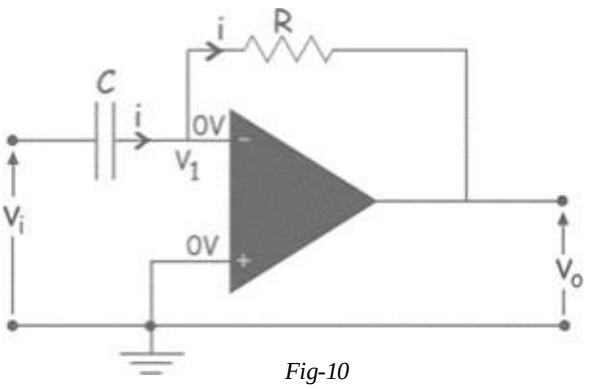


Fig-10

$$C\frac{(v_i - v_{1)}}{dt} = -\frac{v_o}{R}$$

$$C\frac{(v_i - 0)}{dt} = -\frac{v_o}{R}$$

$$C\frac{dv_i}{dt} = -\frac{v_o}{R}$$

$$\therefore \boldsymbol{v_o = -RC\frac{dv_i}{dt}}$$

- The above equation shows that the output voltage is the derivative of the input voltage

7. Difference Amplifier (Subtractor):

- A differential amplifier (also known as a difference amplifier or op-amp subtractor) is a type of electronic amplifier that amplifies the difference between two input voltages but suppresses any voltage common to the two inputs as in fig.

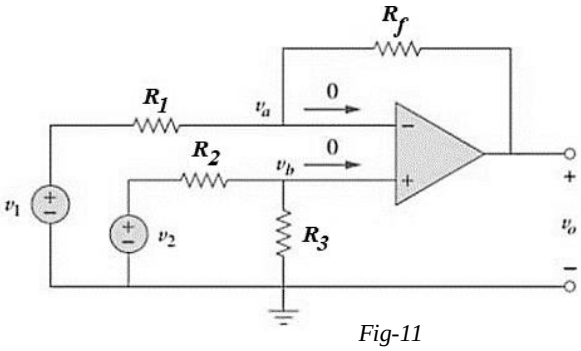


Fig-11

- In the fig-11, applying KCL to node a,

$$\frac{V_1 - V_a}{R_1} = \frac{V_a - V_o}{R_f}$$

$$=> V_o = \left(\frac{R_f}{R_1} + 1\right) V_a - \frac{R_f}{R_1} V_1 \qquad \text{——— (1)}$$

- Applying KCL to node b,

$$\frac{V_2 - V_b}{R_2} = \frac{V_b - 0}{R_3}$$

$$V_b = \frac{R_3}{R_2 + R_3} V_2 \qquad \text{——— (2)}$$

- But summing point $v_a = v_b$ & substituting eqn(2) into eqn(1) yields

$$V_o = \left(\frac{R_f}{R_1} + 1\right) \frac{R_3}{R_2 + R_3} V_2 - \frac{R_f}{R_1} V_1$$

Rearranging

$$V_o = \frac{(1 + R_1/R_f)}{R_1(1 + R_2/R_3)} V_2 - \frac{R_f}{R_1} V_1 \qquad \text{——— (3)}$$

- Since a difference amplifier must reject a signal common to the two inputs, the amplifier must have the property that $v_o = 0$ when $v_1 = v_2$. This property exists when

$$\frac{R_1}{R_f} = \frac{R_2}{R_3}$$

- Then cqn(3) becomes

$$V_o = \frac{R_f}{R_1}(V_2 - V_1)$$

- If $R_f = R_1$ and $R_2 = R_3$, the difference amplifier becomes a *subtractor*, with the output

$$V_o = (V_2 - V_1)$$

Module-3: Communication Systems & Modulation Schemes

I. Modern Communication system scheme:

Communication:

- Transfer of information from one point to other (or) Exchange of Information between two points.

Electronic Communication:

- Sending data or Information using electronic devices. This allows rapid transfer of goods, money and ideas

Need for Communication:

- Speedy transmission: Requires only few seconds to communicate through electronic-media due to the technology available for quick transmission
- Wide Coverage: The whole world has become a global village and communication¬around the globe requires just a second
- Low Cost: Cost of an SMS is cheaper than sending a letter by post-
- Exchange of feedback: Instant exchange of feedback.
- 24/7 accessibility: Can be accessed anytime

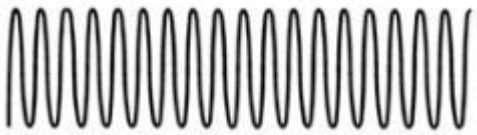
Message signal:

- The signal that consists information is called modulating signal. The modulating signal is usually of low-frequency signal. (20 Hz – 20 KHz)

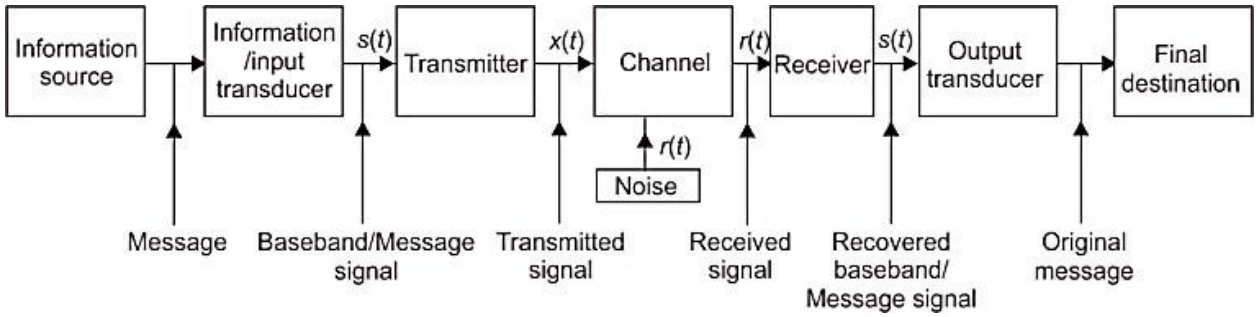


Carrier signal:

- The signal which carries the information is called carrier signal. Carrier signal is a high frequency signal which does not contains any information. The purpose of this signal is just to carry the message signal. The range is in terms of Mega Hertz.



II. General form of a Basic communication system



Fig(1): Schematic block diagram of a basic communication system in most general form

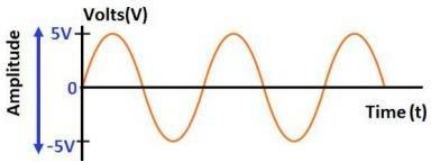
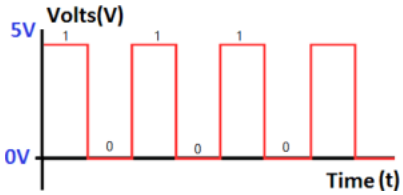
- The main constituents of basic communication system are
  1. Information source and input transducer
  2. Transmitter

3. Channel or medium
4. Noise
5. Receiver
6. Output transducer and final destination

- A communication system transmits information from an information source (message) to a destination. Examples: Voice, Live scenes (video), music, written text, and e-mail.
- A transducer is a device that converts a physical signal into its corresponding electrical signal and vice versa.

1. Information Source and Input Transducer

- Message or information originates in the information source may be in the form of sound (human speech), picture (image source), words (text). However, out of these messages, only the desired message is selected and communicated.
- Type of Signals: Signals are functions that carry information. These are used to convey information from place to place. In electronics, signals are mainly in the form of varying voltages. There are two types of signals.

Analog signal	Digital signal
	
Analog signal is continuous and time varying	Digital signal have two or more states (binary form)
Troubleshooting is difficult.	Troubleshooting is easy.
Easily affected by the noise.	These are stable and less prone to noise
Analog signals use continuous values to represent the data, usually in the form of sine wave	Digital signals use discrete values to represent the data, usually in the form of square wave
Accuracy may be affected by noise.	Accuracy is immune from the noise
Analog signals may be corrupted during data Transmission	Digital signals are not corrupted during data transmission
Analog signals use more power.	Digital signals use less power.
Examples: Temperature, Pressure, Flow measurements etc.	Examples: Motor Start, Trip, etc
Components like resistors, Capacitors, Inductors, Diodes are used in analog circuits	Components like transistors, logic gates, and micro-controllers are used in digital circuits

- A transducer is a device which converts one form of energy into another form. Generally, the input transducer converts the non-electrical signal (Ex: sound signal or light signal) into an electrical signal.
- For example, in case of radio-broadcasting, a microphone converts the information or massage which is in the form of sound waves into corresponding electrical signal.

2. Transmitter

- The base band signal (electrical form), output from the input transducer is applied to the input of the transmitter. The transmitter section processes the signal prior to transmission. The nature of processing depends on the type of communication system. There are two options for processing signals prior transmission
  - (i)The baseband signal, which lies in the low frequency spectrum, is translated to a higher frequency spectrum --- carrier communication system. In this process, modulation is the main

function of the transmitter. In modulation, the message signal is superimposed upon the high-frequency carrier signal. As the original message signal cannot be transmitted over a longer distance because of their low frequency and amplitude, they are superimposed with high frequency carrier signal.

- (ii)The baseband signal is transmitted without translating it to a higher frequency spectrum --- baseband communication system

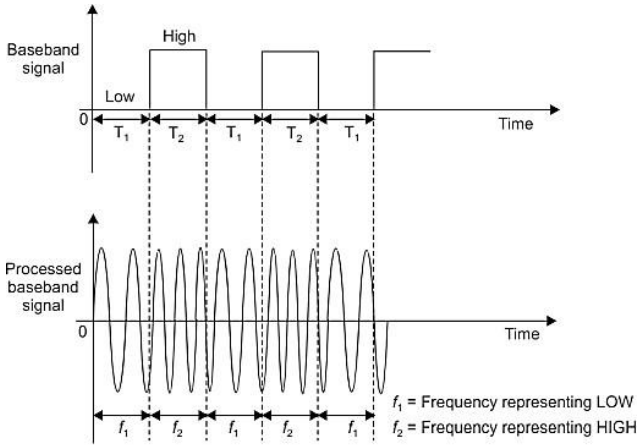


Fig. The processing of a baseband signal

- If the signal is to be transmitted through the space, then the transmitter will convert the electrical signals into radio waves. If the signal is to be transmitted through the fiber optic cable then the transmitter will convert the electrical signal into light or optical signal.

2.1. Typical Analog transmitter

- Let s(t) be the baseband signal applied to the modulated stage as shown in the fig1, to translate (modulate) it from low to high frequency spectrum. That means, s(t) is superimposed upon the high-frequency carrier signal c(t), which is generated by a high frequency carrier oscillator. The output of the modulated stage is called the modulated signal, and is designated as x(t). This signal is then applied to amplifier stage. The voltage of the modulated signal is amplified to drive the power amplifier stage. The power amplifier that amplifies the power of the modulated signal x(t) and thus it carries enough power to reach the receiver stage of the communication system. Finally, the signal is passed to the transmission medium or channel.

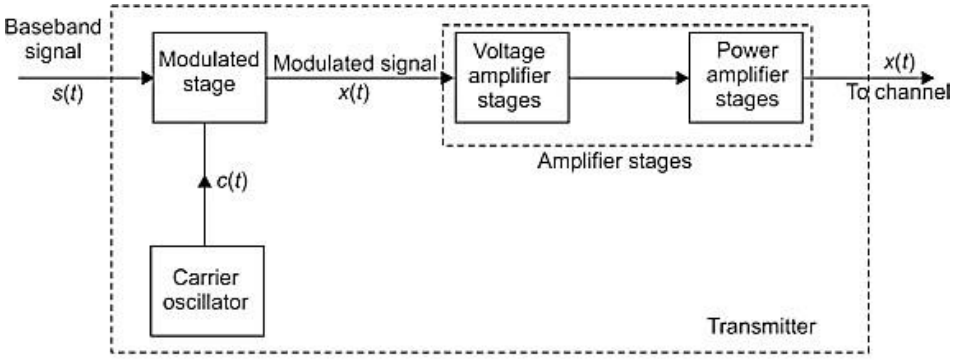


Fig. Block diagram representing schematic of an analog transmitter section

- For example, in radio broadcasting the sound signal (message signal) is converted into the electrical signal (baseband signal), by the input transducer. Then, it is modulated with the Radio Frequency (RF) range carrier signals to produce radio signals. These radio signals are transmitted through electromagnetic waves (open space), referred to as radio waves.

- RF bands spread in the range between 30 kHz and 300 GHz. Transmission systems are operated in the RF spectrum range including analog radio, aircraft navigation, marine radio, amateur radio, TV broadcasting, mobile networks and satellite systems.

Frequency Name	Frequency Range	Application
Super high frequencies	3GHz-30GHz	RADAR
Ultra high frequencies	300MHz-3GHz	Satellite communication, cellular phones
Very high frequencies	30MHz-300MHz	TV and FM
High frequencies	3MHz-30MHz	Commercial short wave broadcast
Medium frequencies	300kHz-3MHz	AM radio broadcasting, navigational systems, maritime communication, air traffic control
Low frequencies	30kHz-300kHz	Navigation, submarine communication
Very low frequencies	3kHz-30kHz	Navigation, submarine communication
Voice frequencies	300Hz-3kHz	Navigation ,submarine, audio
Extremely low frequencies	30Hz-300Hz	Power transmission

Table 1: Frequency ranges and its application in communication systems

3. Channel or medium:

- The term channel means the medium through which the message travels from the transmitter to the receiver. The transmitted signal should have adequate power to withstand the channel noise. The channel characteristics also impose constraints on the bandwidth. Depending on the physical implementations, one can classify the channels in the following two groups:

- **3.1. Hardwired channels** (Manmade structure) Transmission lines: It is a conductive medium consisting of two or more conductors through which electrical signal are transmitted from transmitting point to receiver point.

  Example:

  - Twisted pair cables used in telephony, in which two conductors are twisted together for the purposes of improving electromagnetic compatibility.
  - Coaxial cable used in TV transmission, to carry high-frequency electrical signals with low losses.
  - Waveguide: consisting of a hollow, metal tube of uniform cross-section used for transmitting electro- magnetic waves. When signals entered the waveguides are reflected at the metallic walls and propagate at the other end. The energy is in the form of electric field and magnetic field which are perpendicular to each other.
  - Optical Fibre: consist of very thin hollow glass fibre through which signal is transmitted in the form of light energy.

- **3.2. Soft-wired channels** (no physical link between transmitter and receiver): Natural resources which can be used as the transmission medium for signals.

  Example: Air or Open space and Sea water. The signals are transmitted in the form of electromagnetic (EM) waves also called radio waves. Radio waves travel through open space at a speed equal to that of light (c = 3 x 108 m/s)

4. Noise:

- Noise is defined as unwanted electrical signal which do not have any useful information. Noise is a highly undesirable part of a communication system, and has to be minimized. When noise is mixed with the transmitted signal, it rides over it and deteriorates its waveform.

4.1. Signal to Noise Ratio (SNR) and Noise figure (F)

- SNR is defined as the ratio of signal power (S) to the noise power (N), often expressed in decibels (dB).

If P_s is signal power and P_n is noise power, then SNR expressed as S/N, is given as

$$\frac{S}{N} = \frac{P_s}{P_n}$$

If $P_s = V_s^2 R$ and $P_n = V_n^2 R$, then

$$\frac{S}{R} = \frac{P_s}{P_n} = \frac{V_s^2 R}{V_n^2 R}$$

where V_s is signal voltage and V_n is noise voltage.

In addition, it is assumed that both the signal and noise powers are dissipated in the same resistor R. Therefore, SNR can be expressed in terms of decibels (dB) as

$$\left(\frac{S}{N}\right)_{dB} = 10 \log_{10}\left(\frac{V_s^2}{V_n^2}\right)$$

$$\left(\frac{S}{N}\right)_{dB} = 20 \log_{10}\left(\frac{V_s}{V_n}\right)$$

For example, if, at a particular point in a circuit, the signal and noise voltages are given as 3.5 mV and 0.75 mV, respectively, SNR in dB is calculated as:

$$\left(\frac{S}{N}\right)_{dB} = 20 \log_{10}\left(\frac{3.5}{0.75}\right)$$

or $$\text{SNR} = 20 \log_{10}(4.66) = 13.38\ dB$$

Noise Factor, *F*

$$F = \frac{Si / Ni}{So / No} = \frac{\text{SNR at input}}{\text{SNR at output}}$$

Noise Figure, *NF*

$$NF = 10 \log F$$
$$= 10 \log \frac{Si / Ni}{So / No}\ (dB)$$

5. Receiver:

- The main function of the receiver is to reproduce the original message signal. This reproduction of the original signal is accomplished by a process known as the demodulation or detection. Demodulation is the reverse process of modulation carried out in transmitter. The signal received by the receiver is r(t). Due to attenuation this received signal r(t) is a weak signal. A voltage amplifier amplifies to make strong enough for further processing

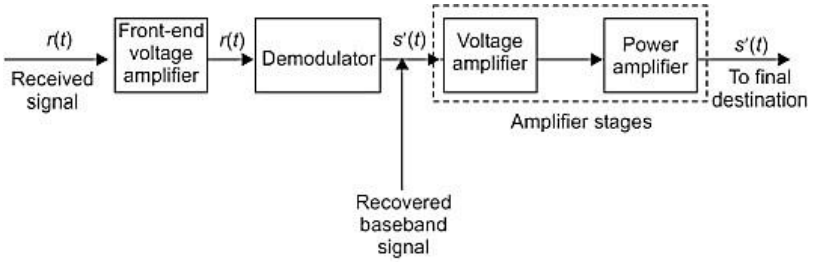


Fig. Detailed block diagram of a typical receive section

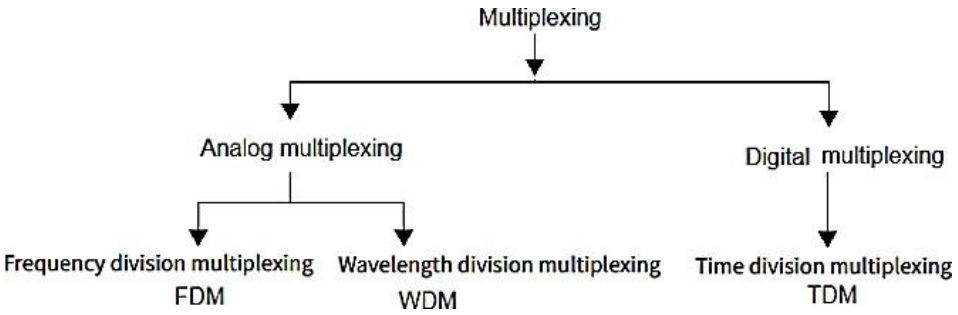
- Next, this signal is applied to the demodulator. In demodulation, the baseband signal is separated from the high-frequency carrier signal. After recovering the original baseband signal s' (t), its voltage and power is amplified before send to the final destination block.

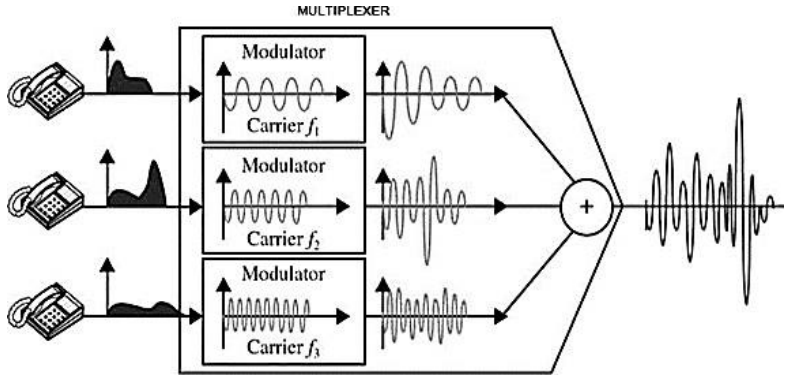
6. Destination:

- Destination is the final stage which is used to convert an electrical message signal into its original form. For example in radio broadcasting, the destination is a loudspeaker which works as a transducer that converts the electrical signal to original sound signal.

III. Multiplexing:

- Multiplexing is a process which allows more than one signal to be transmitted through a single channel.





- In multiplexing, each baseband signals are modulated with different carrier frequencies. At the transmitter they can be mixes and transmitted. At the receiver end, all different transmitted signals can be easily separated by the known carrier frequencies.
- Due to multiplexing it is possible to increase the number of communication channels so that more information can be transmitted where it makes the communication system economical. The typical applications of multiplexing are telephone, satellite communication etc.

IV. Types of Communication Systems:

I. Communication Systems based on Physical Infrastructure:

- Line communication systems: Uses power lines to transfer data from one point to another point. There is a physical link, called a hardwire channel between the transmitter and the receiver inline communication systems. Ex: Land line telephony, Cable TV.

II. Communication systems based on Signal specifications:

- **A. Based on Nature of baseband or information signal**
    - (i) Analog communication systems: Exchange of information between two points through analog signals. Ex: Audio, video and pictures between two points using the analog signals.

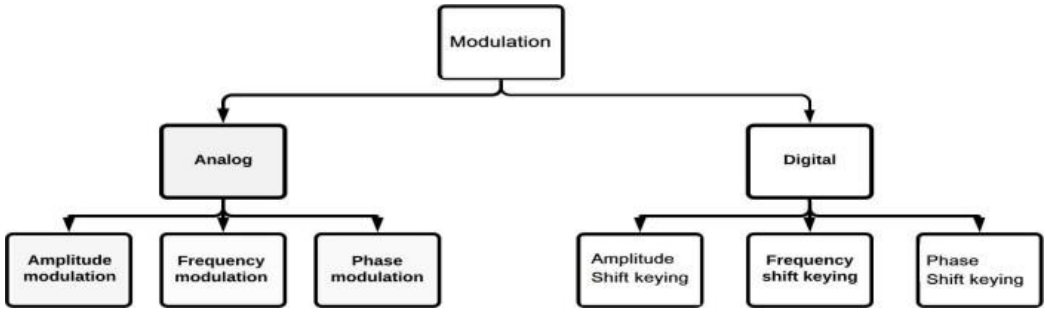
- (ii) Digital communication systems: Exchange of information between two points through digital signals. Ex: Audio, HDTV

➢ **B. Based on Nature of the transmitted signal**

- (i)Baseband communication system: Baseband signals are transmitted without translating (or amplified) to higher frequencies. Ex: Land line, fax, etc
- (ii)Carrier communication system: The baseband signal (low frequency) is mixed with high frequency carrier signal. Ex: Radio, voice messages and calls.

V. Types of Modulation:

➢ Modulation is the process in which any one of the parameters (amplitude, frequency or phase) of the high frequency carrier signal is varied according to the instantaneous values of the low frequency message signal, keeping other parameters constant.



➢ Analog modulation is typically used for AM, FM radio, and short-wave broadcasting. Digital modulation involves transmission of binary signals (0 and 1).

1. Amplitude Modulation (AM):

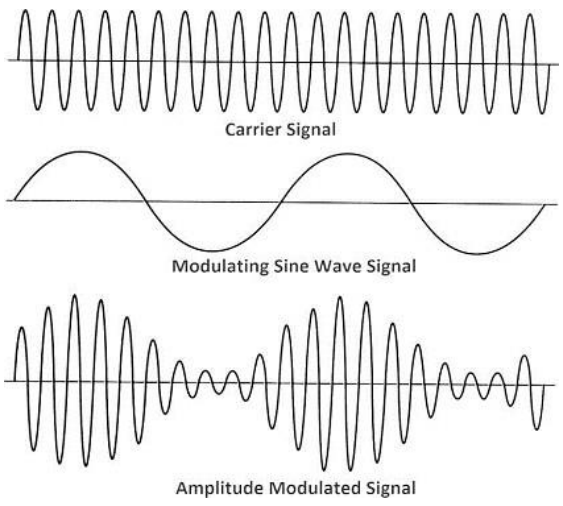
➢ It is the process in which the amplitude of the carrier signal is varied according to the instantaneous values of the message signal, whereas the frequency and phase are kept constant. It is as shown in the fig.

➢ The first figure is the carrier wave, which is a high frequency signal and contains no information. Denoted as $c(t) = Ac \cos(2\pi fct)$

➢ The next one is the modulating wave, which is the message signal; (low frequency signal) contains information. Denoted as $m(t) = Am \cos(2\pi fmt)$

➢ The last one is the resultant amplitude modulated wave.

➢ Mathematically, AM is denoted as

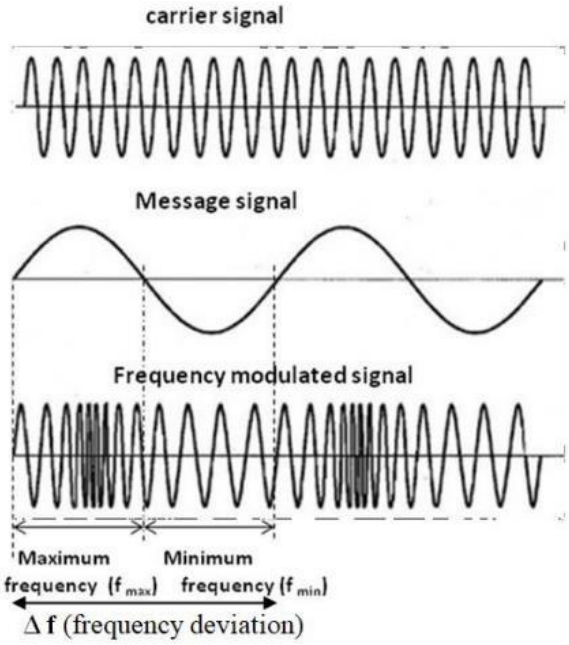


$$V_{AM} = V_c \sin \omega_c t + \frac{m_a V_C}{2} \cos (\omega_c - \omega_m) t - \frac{m_a V_C}{2} \cos (\omega_c + \omega_m) t$$

➢ In AM, most of the transmitted power is wasted in carrier, but used for longer distance communication.

2. Frequency Modulation (FM):

➢ It is defined as a process in which the frequency of the carrier is varied in accordance with the instantaneous values of the message signal, whereas the amplitude and phase are kept constant.



Mathematically, FM wave is denoted as $S(t) = A \sin [2\pi f_c t + m_f \sin (2\pi f_m t)]$

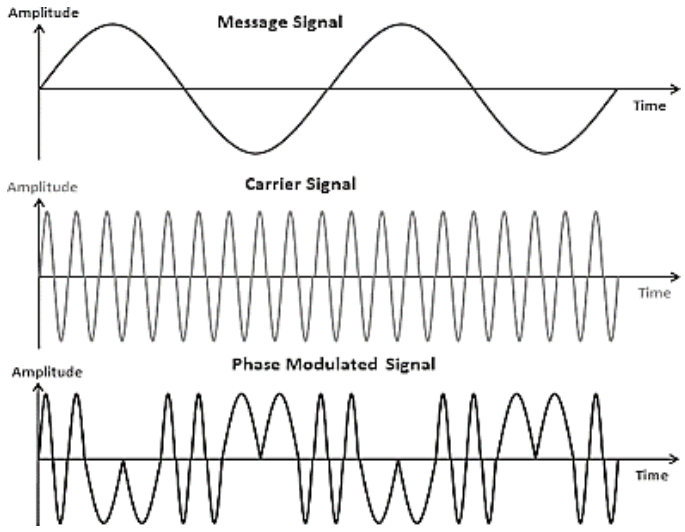
Modulation index (depth of modulation) $m_f = \frac{\Delta f}{f_m}$

Carrier Swing = $f_{(max)} - f_{(min)} = 2\Delta f$

- In FM, all the transmitted power is useful, but used for short distance communication.

3. Phase Modulation:

- It is defined as a process in which the phase of the carrier is varied in accordance with the instantaneous values of the message signal, whereas the amplitude and frequency are kept constant.



Mathematical equation of PM wave will be $s(t)=A_c \cos(2\pi f_c t+\beta\cos(2\pi f_m t))$

Where, β = ***modulation index*** = $\Delta\phi = k_p A_m$ and $\Delta\phi$ is phase deviation.

VI. Radio Wave Propagation:

- Radio waves exhibit the properties of light with velocity 3x108m/s. These are electromagnetic waves that consist of electric and magnetic field components. It is traversed in nature. Radio signals can travel from one end to another over vast distances. Since these are EM waves, they exhibit properties (like light waves) such as reflection, refraction, diffraction, absorption, polarization and scattering.
- Radio propagation is the way of transmitting radio signals in different ways:

1. Ground or surface wave:

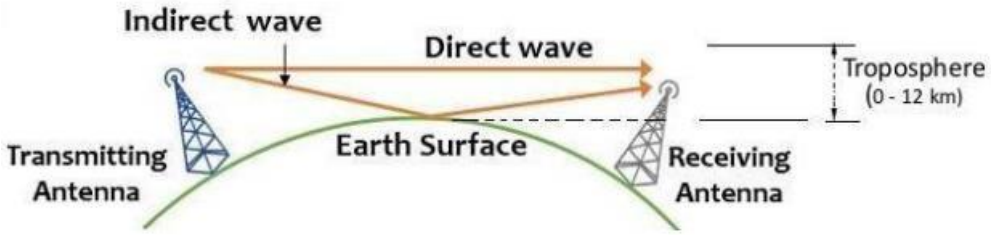
- Ground waves can be used for radio communication. Ground wave transmission is very reliable irrespective of the atmospheric conditions.



- Frequency range: 30 kHz to 3 MHz Transmission distance: 100 to 1000 km Example: AM radio broadcast in the medium frequency band cover local areas.

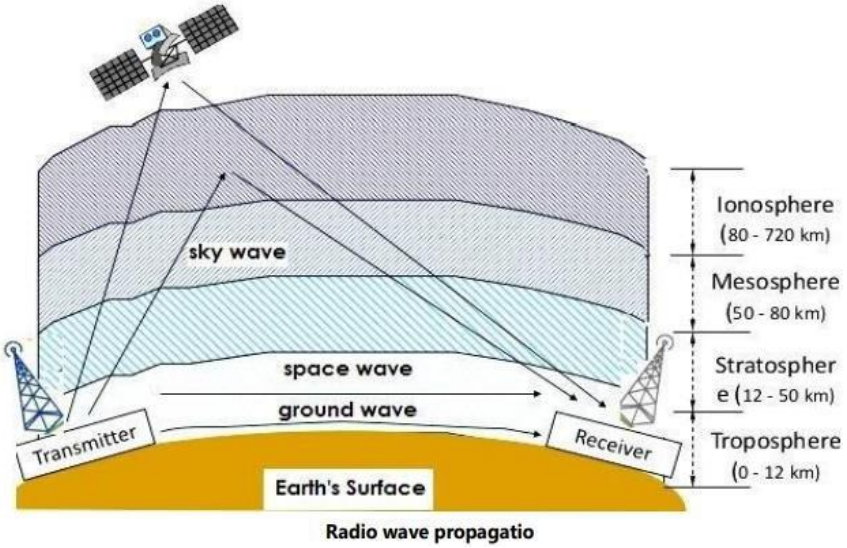
2. Space or tropospheric wave

- In space wave (or line of sight propagation), radio waves move in the earth's troposphere within about 12 KM over the surface of the earth. Frequency range: 3MHz to 30 MHz. Example, TV Transmission. The space wave is made up of two components: (a)a direct or line-of- sight wave from the transmitting to the receiving antenna and (b)an indirect or ground-reflected wave traversing form the transmitting antenna to ground and reflected to the receiving antenna



3. Sky wave

- Radio waves transmitted from the transmitting antenna reach the receiving antenna after reflection from the ionosphere (the earth's upper atmosphere). Sky wave is responsible for short wave transmission around the globe via successive reflections at the ionosphere and the earth's surface. Ionosphere - The ionized region extending about 80 KM above the earth's surface. In ionosphere radiation from the sun ionizes atoms and molecules that liberate electrons and ions from molecules. The propagation of radio wave through the ionosphere is affected by the electrons and ions. The effect of the electrons on the propagation is much greater than that of the ions since the electronic mass is much less than the ionic mass.



Radio wave propagatio

- Advantages of sky wave propagation

- It supports large distance propagation.
- The frequency range of operation is considerably high.
- Attenuation due to atmospheric conditions is less.

➢ Disadvantages of sky wave propagation
- Long-distance propagation requires large-sized antennas.
- Due to the presence of the ionosphere near and far during night and day respectively there exist variation in signal transmission in day and night.

➢ Applications
- Sky wave propagation is widely used in mobile and satellite communications

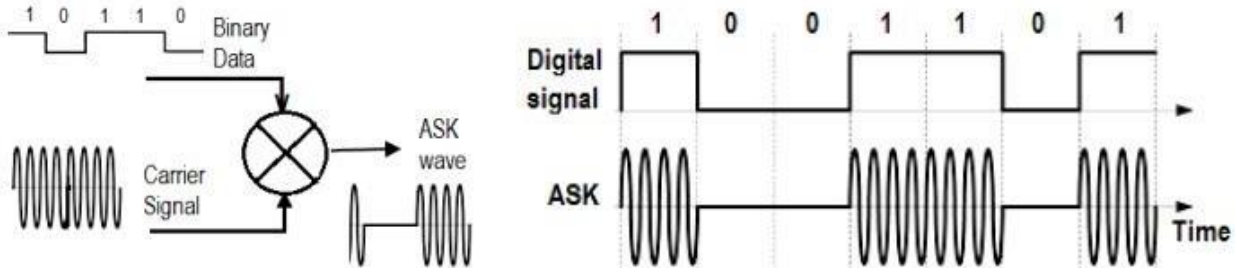
VI. Digital Modulation Schemes:

➢ In digital communications, the modulating signal consists of binary data. When it is required to transmit digital signals, the amplitude, frequency or phase of the sinusoidal carrier is varied in accordance with the incoming digital data.

➢ Digital modulation schemes are classified as
1. Amplitude Shift Keying (ASK)
2. Frequency Shift Keying (FSK)
3. Phase Shift Keying (PSK)

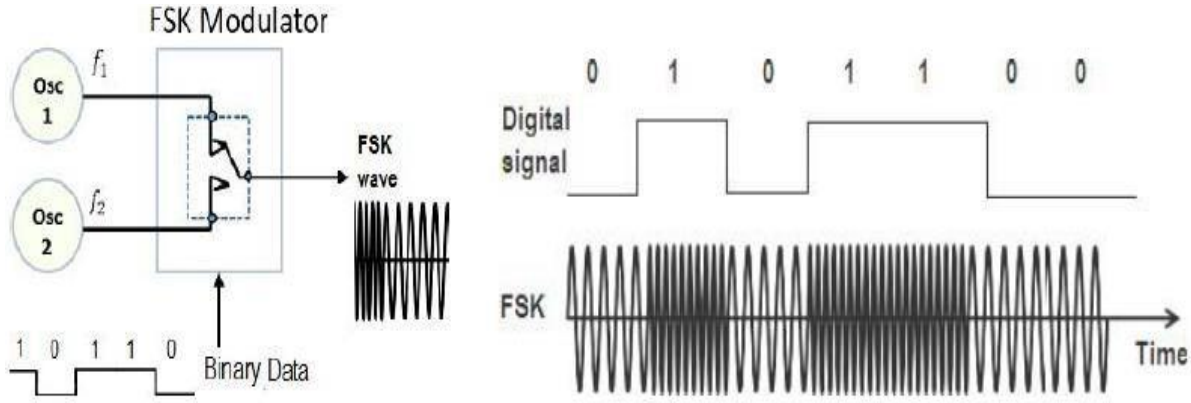
1. ASK (Amplitude Shift Keying):

➢ ASK represents digital data as variations in the amplitude of a carrier wave. ASK signal can generated when the incoming binary data and the sinusoidal carrier are applied a product modulator as inputs.



2. FSK (Frequency Shift Keying):

➢ In this technique digital signal is transmitted by switching between low frequency and high frequency in order to represent 0's and 1's. The simplest FSK is Binary FSK (BFSK). It uses a pair of discrete frequencies to transmit binary (0s and 1s) information.



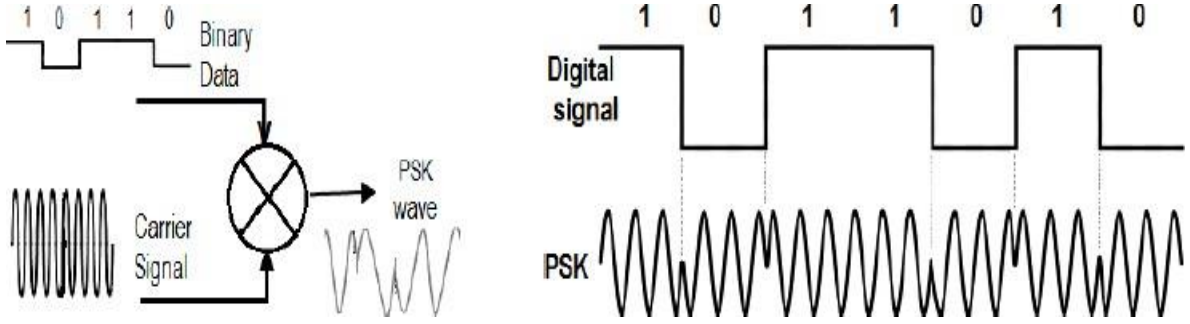
3. Phase Shift Keying (PSK):

- The simplest form of PSK is binary phase shift keying (BPSK). In this case, the carrier phase is shifted between two different phases (typically 0° and 180°) depending on whether 0-bit or 1-bit is being transmitted.

For example:

0-bit: the symbol transmitted is $Vc \cos(2\pi f_c t)$

1-bit: the symbol transmitted is $Vc \cos(2\pi f_c t + 180°) = -Vc \cos(2\pi f_c t)$



VII. Radio Signal Transmission:

- Important components of wireless transmission system

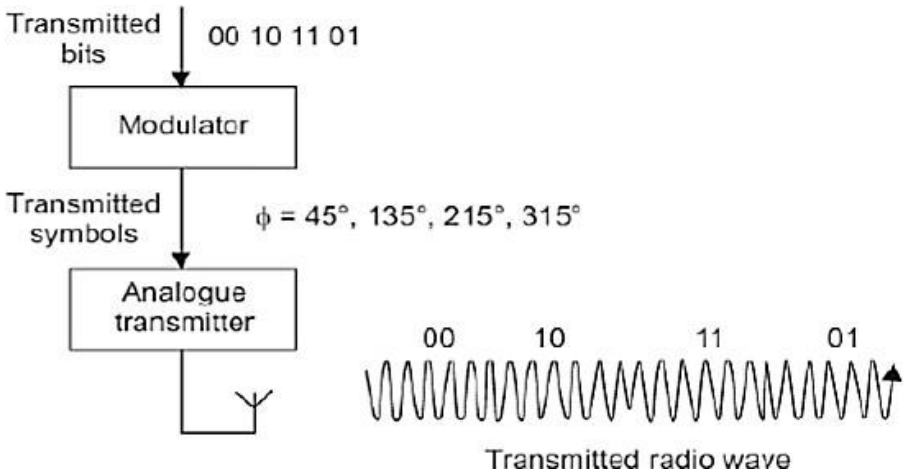


Fig: Architecture of wireless communication transmitter

- The transmitter accepts a stream of bits from the application software and encodes these bits onto a radio wave, known as a carrier, by adjusting parameters of the wave such as its amplitude or phase.
- Transmitter processes the incoming bits in two stages

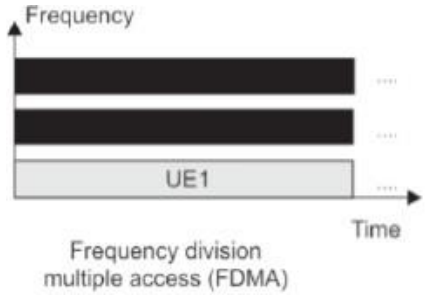
  1. Modulator – accepts the incoming bits and computes symbols that represent the amplitude and phase of the outgoing wave. Modulation scheme used –QPSK – the modulator takes 2 bits at a time and transmits them using radio waves having 4 different states with phases of 45^0, 135^0, 225^0 and 315^0.
  2. Analogue transmitter – Generates the radio wave

VIII. Multiple Access Techniques:

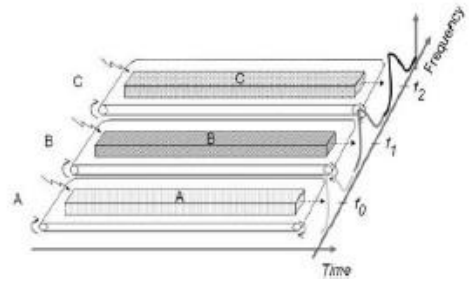
- In cellular network, the base station has to transmit to many different mobiles at once by sharing the resources of air interface using a technique known as multiple access.
- Multiple access techniques
  1. Frequency Division Multiple Access (FDMA)
  2. Time Division Multiple Access (TDMA)
  3. Code Division Multiple Access (CDMA)

1. Frequency Division Multiple Access (FDMA):

- Each mobile receives signal on its own carrier frequency, which it distinguishes using¬ analogue filter. The carriers are separated by unused guard rings to minimize the interference¬ between the carriers.

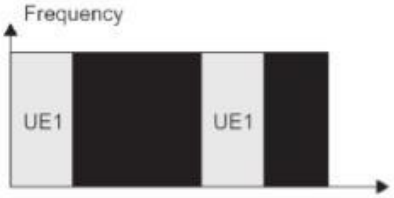


Frequency division multiple access (FDMA)

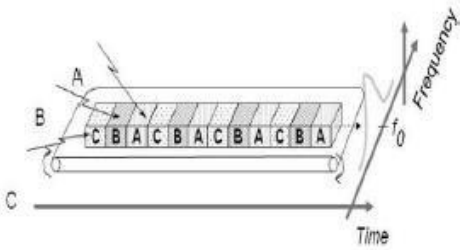


2. Time Division Multiple Access (TDMA):

- Mobiles receive signals on the same carrier but at different times.

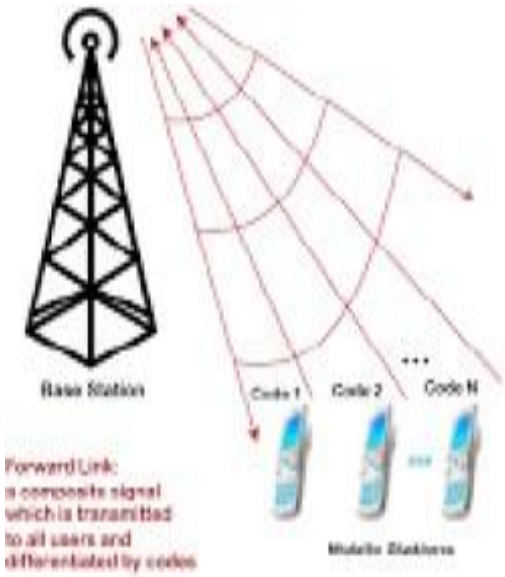


Time division multiple access (TDMA)



3. Code Division Multiple Access (CDMA):

- Technique used by third generation communication systems
- Mobiles receive signals on the same carrier frequency and at the same time.
- The signals are labelled by the use of codes, which allows a mobile to separate its own signal from the others.





Module-4: Embedded Systems

2. Embedded System

- An embedded system is an electronic/electro-mechanical system designed to perform a specific function and a combination of both hardware and firmware (software).
- Every embedded system is unique and the hardware as well as the firmware is highly specialized to the application domain.
- Embedded systems are becoming an inevitable part of any product or equipment in all fields including household appliances, telecommunications, medical equipment, industrial control, consumer products, etc.
- Embedded system is a combination of 3 things
  1. Hardware
  2. Software
  3. Mechanical component & it is supposed to do only one specific task.

Examples:

Example1: Washing Machine

- A washing machine from an embedded systems point of view has:
  - a. Hardware: Buttons, displays & buzzer, electronic circuitry.
  - b. Software: It has a chip on the circuit that holds the software which drives controls & monitors various operations possible.
  - c. Mechanical components: the internals of a washing machine which actually wash the clothes control the input and output of water.

Example-2: Air Conditioner

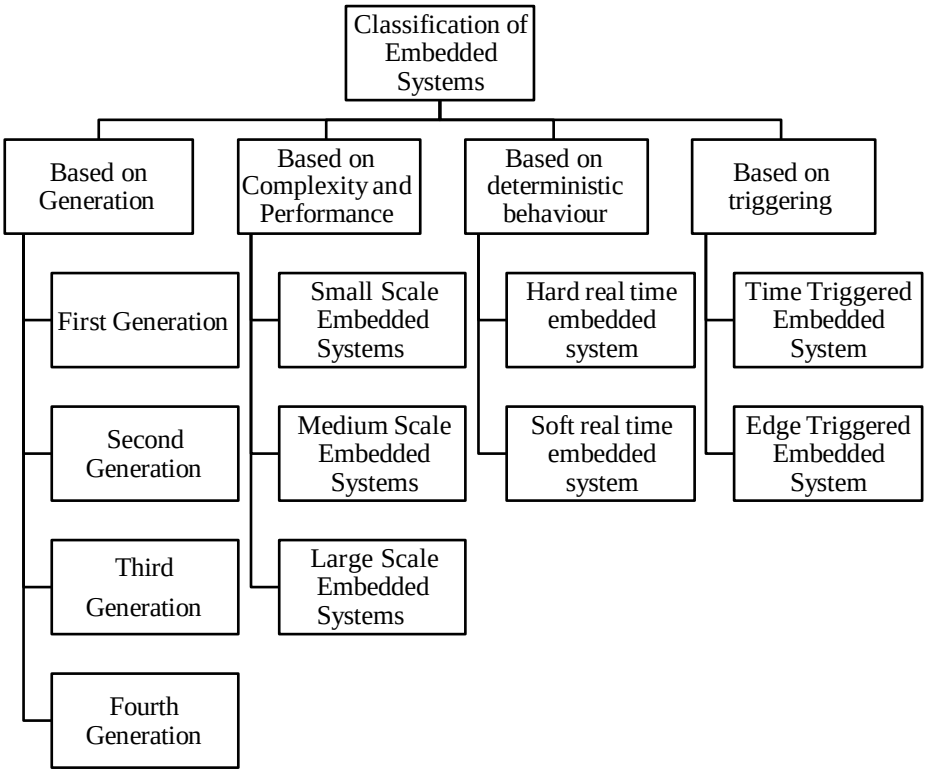
- An Air Conditioner from an embedded systems point of view has:
- Hardware: Remote, display & buzzer, infrared Sensors, electronic circuitry
- Software: It has a chip on the circuit that holds the software which drives control & monitors the various operations possible. The software monitors the external temperature through the sensors and then releases the coolant or suppresses it.
- Mechanical components: The internals of an air conditioner the motor, the outlet, etc.-

3. Differences between General Purpose computing system and Embedded system

Criteria	General Purpose computing system	Embedded system
Contents	A system which is a combination of a generic hardware and general-purpose operating system for executing a variety of applications	A system which is a combination of a special-purpose hardware and embedded OS for executing a variety of appli
OS	It contains a general-purpose operating system (GPOS)	It may or may not contain an operating system for functioning
Alterations	Applications are alterable (programmable) by the user. (It is possible for end user to re-install the OS and also add or remove user applications)	The firmware of the Embedded system is pre-programmed and it is non-alterable by the end user.
Key Factor	Performance is the key deciding factor in the selection of the system. Faster is better.	Application specific requirements (like performance, power requirements,

		memory usage etc) are the key deciding factor. Power
Power consumption	More	Less
Response Time	Not Critical	Critical for some applications
Execution	Need not be deterministic	Deterministic for certain types of ES like 'Hard real time systems.

4. Classification of Embedded System:



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1. Based on Generation

1. a. First Generation

- The earlier first-generation embedded systems were built around 8-bit microprocessors and 4-bit microcontrollers. Such embedded system possesses simple hardware and firmware developed using assembly code.
- Ex: Digital telephone keypads, stepper motor control units.

1. b. Second Generation

- After the evolution of the second generation embedded systems, the 8-bit processor and 4-bit controllers are replaced by 16-bit microprocessors and 8-bit microcontrollers. They are more powerful and complex compared to previous generation processors.
- Ex: Data acquisition systems, SCADA systems.

1. c. Third Generation

- Embedded Systems built around high performance 32-bit microprocessors and 16-bit microcontrollers. Hence, its operation has become much more powerful and complex than the second generation.

- During this period, domain-specific processors/controllers like Digital Signal Processors (DSP), Application- Specific Integrated Circuits (ASICs) and the concept of instruction pipelining, embedded real-time operating system evolved into the embedded system industry.
- Ex: Robotics, industrial process control, embedded networking.

1. d. Fourth Generation

- Embedded Systems built around System on Chips (SoCs), Re-configurable processors and multi-core processors, coprocessors also emerged into the embedded market to add more powerful performance
- These systems also make use of the high-performance real-time.

2. Based on Complexity and Performance

2. a. Small Scale Embedded Systems

- Small Scale Embedded Systems are built with a single 8 or 16-bit microprocessor or controller. The main programming tools used are an editor, assembler, cross assembler and integrated development environment (IDE).
- The hardware and software complexities in small-scale embedded system are very low. It may or may not contain an operating system for its functioning. An electronic toy is an example for a small-scale embedded system.

2. b. Medium Scale Embedded Systems

- The Embedded system with medium performance 16-bit or 32-bit microprocessor or controller, ASICs or DSPs fall under the medium scale embedded systems.
- They have both hardware and software complexities The main programming tools used are C, C++, JAVA, Visual C++, RTOS, debugger, source code engineering tool, simulator and IDE.

2. c. Large Scale Embedded Systems

- The embedded systems have highly complex hardware and software, built around 32-bit or 64-bit processors/controllers, RISC processors, SoC, scalable and configurable processors. They are also called sophisticated embedded systems.
- They are used for cutting-edge applications that need hardware and software co-design, where components have to be assembled into the final system. They also contain a high-performance real-time operating system for task scheduling, prioritization and management.

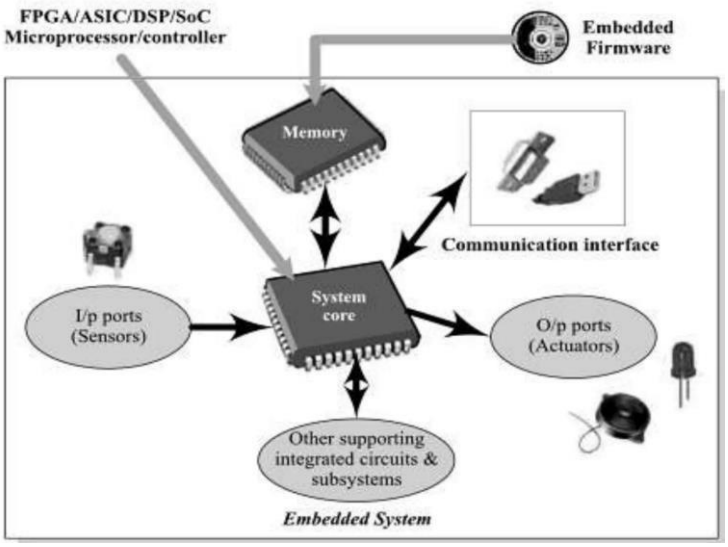
5. Major application areas of embedded systems

The application areas and the products in the embedded domain arc countless. A few of the important domains and products are listed below:

1. Consumer electronics: Camcorders, cameras, etc.
2. Household appliances: Television, DVD players, washing machine, fridge, microwave oven. etc.
3. Home automation and security systems: Air conditioners, sprinklers, intruder detection alarms, closed circuit television cameras, fire alarms
4. Automotive industry: Anti-lock breaking systems (ABS), engine control, ignition systems, automatic navigation systems,
5. Telecom: Cellular telephones, telephone switches, handset multimedia applications, etc.
6. Computer peripherals: Printers, scanners, fax machines, etc.
7. Computer networking systems: Network routers, switches, hubs, firewalls, etc.
8. Healthcare: Different kinds of scanners, EEG, ECG machines etc.

9. Measurement & Instrumentation: Digital multi meters, digital CROs, logic analysers PLC systems, etc.
10. Banking & Retail: Automatic teller machines (ATM) and currency counters, point of sales (POS)
11. Card Readers: Barcode, smart card readers, hand held devices, etc.

6. Elements of Embedded Systems



A typical embedded system contains a single chip controller, which acts as the master brain of the system.

- The controller can be a Microprocessor (e.g. Intel 8085) or a microcontroller (e.g. Atmel AT89C51) or a Field Programmable Gate Array (FPGA) device (e.g. Xilinx Spartan) or a Digital Signal Processor (DSP) or an Application Specific
- Embedded systems are basically designed to regulate a physical variable or to manipulate the state of some devices by sending some control signals to the Actuators or devices connected to the O/p ports of the system, in response to the input signals provided by the end users or Sensors which are connected to the input ports. Common user interface input devices whereas LEDs, liquid crystal displays, piezoelectric buzzers, etc. are examples for common user interface output devices for a typical embedded system.
- The Memory of the system is responsible for holding the control algorithm and other important configuration details.
- An embedded system without code implemented memory has all the peripherals but is not capable of making decisions depending on the situational as well as real world changes.
- Memory for implementing the code may be present on the processor or may be implemented as a separate chip interfacing the processor.

7. Core of the Embedded System

The core of the embedded system falls into any one of the following categories:

1. General Purpose and Domain Specific Processor
   1.1 Microprocessors
   1.2 Microcontrollers
   1.3 Digital Signal Processors
2. Application Specific Integrated Circuits (ASICs)
3. Programmable Logic Devices (PLDs)
4. Commercial off-the-shelf Components (COTS)

Microprocessor	Microcontroller
Consists of a CPU, performs Arithmetic and Logical operations. Ex: Intel 8086 microprocessor	Highly integrated chip contains CPU, RAM, on chip ROM/flash memory, I/O ports. Ex: Intel 8051 microcontroller
It is mainly used in Personal Computers	It is mainly used in an embedded system
Complex and expensive, with a large number of instructions to process.	Simple and inexpensive with less number of instructions to process.
Dependent Unit	Self-contained unit
Consumes more power	Consumes less power
Limited power saving options	Includes lot of power saving features
Architecture is based on Von Neumann model	Architecture is based on Harvard architecture
Uses an external bus to interface to RAM, ROM, and other peripherals	Uses an internal controlling bus.

RISC vs. CISC

RISC	CISC
Reduced Instruction Set Computer	Complex Instruction Set Computer
Software centric design.	Hardware centric design.
Low power consumption.	High power consumption.
Requires more RAM	Requires a minimum amount of RAM
Simple decoding of instruction	Complex decoding of instruction.
Processors are highly pipelined	Processors are not pipelined or less pipelined
Execution time is very less	Execution time is very high
Uses multiple registers	Uses a single register.
It does not require external memory for calculations	It requires external memory for calculations
Compound addressing mode.	Limited addressing mode.
RISC architecture can be used with high-end applications like telecommunication, image processing, video processing, etc	CISC architecture can be used with low-end applications like home automation, security system, consumer goods etc.
Small Code Size.	Large Code Size.
Fixed Instruction format (32-bit)	Varying formats (16 to 64 bits for each instruction).
xamples: ARM, PIC, Power Architecture, Alpha, AVR, ARC and the SPARC	Examples: VAX, Motorola 68000 family, System/360, AMD and the Intel x86 CPUs.

Harvard vs. Von-Neumann Processor

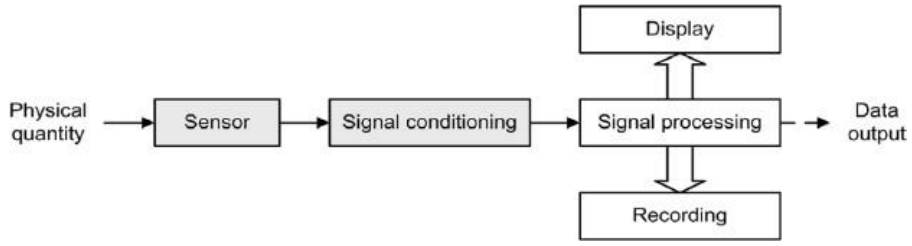
Von-Neumann Architecture	Harvard Architecture
It is ancient computer architecture based on stored program computer concept	It is modern computer architecture based model.
CPU is connected t o data memory (RAM) and program memory (ROM) by a single memory.	CPU is connected t o data memory (RAM) and program memory (ROM), separately
CPU cannot access instructions and data at the same time.	CPU can access instructions and data at the same time.
Same physical memory address is used for instructions and data.	Separate physical memory address is used for instructions and data

Common bus is used for data and instruction transfer.	Separate buses are used for data and instruction transfer.
The speed of execution is slower. It is because it is not capable of fetching the instructions and data both at the same time.	The overall speed of execution is faster. It is because the processor is capable of fetching both instructions and data at the very same time.
It is cheaper	It is costly.
Requires less hardware, but low performance	Requires more hardware, but high performance.
It is used in personal computers and small computers	It is used in microcontrollers and digital signal processing.

8. Sensors and Interfacing

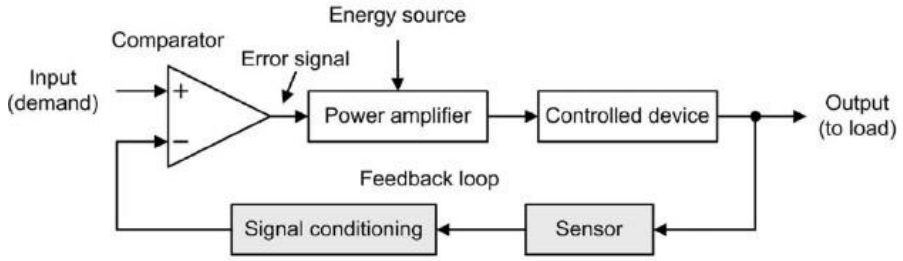
Instrumentation and control systems

- Fig(a) shows the arrangement of an instrumentation system.



- The physical quantity to be measured (e.g. temperature) acts upon a sensor that produces an electrical output signal.
- This electrical signal is passed to the sensor, the output produce by sensor contain unwanted signals(ie noise),so further signal conditioning is required.
- Once the signal is at an acceptable level and in an acceptable form for signal processing, display and recording.
- Furthermore, because the signal processing may use digital rather than analogue signals an additional stage of analogue-to-analogue conversion may be required.

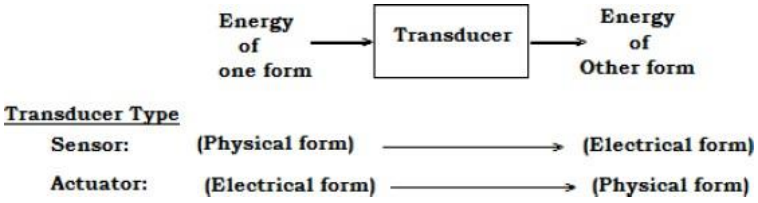
- Fig (b) shows the arrangement of a control system.



- This uses negative feedback in order to regulate and stabilize the output.
- Set the input or demand and leave the system to regulate itself by comparing it with a signal derived from the output.
- A comparator is used to sense the difference in these two signals and where any discrepancy is detected the input to the power amplifier is adjusted accordingly. This signal is referred to as an error signal
- The input is often derived from a simple potentiometer connected across a stable d.c. voltage source while the controlled device can take many forms(e.g. a d.c. motor, linear actuator, heater, etc.).

9. Transducers

- Transducers are devices that convert energy in the form of sound, light, heat, etc., into an equivalent electrical signal, or vice versa.



- Examples:
- A microphone is a transducer converting sound pressure variations into voltage or current.
- A loudspeaker is a transducer that converts low frequency electric current into audible sounds.

10. Sensors

- A sensor is a special kind of transducer that is used to generate an input signal to a measurement, instrumentation or control system.
- Sensors can be categorized as either active or passive. An active sensor generates a current or voltage output. A passive transducer requires a source of current or voltage
- Sensors can also be classed as either digital or analogue. The output of a digital sensor can exist in only two discrete states, either 'on' or 'off', 'low' or 'high', 'logic 1' or 'logic 0', etc.
- The output of an analogue sensor can take any one of an infinite number of voltage or current levels.

11. Actuators

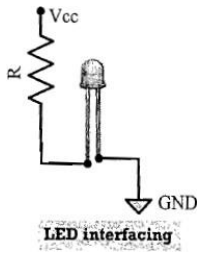
- Actuator is a form of transducer device (mechanical or electrical) which converts signals to corresponding physical action (motion). Actuator acts as an output device.

Sensor	Actuator
It converts physical quantity into electrical signals.	It converts electrical signals into physical quantity.
It takes input from environment.	It takes input from the electric or electronic system.
Sensor generated electrical signals	Actuator generates heat, motion, vibration etc.
It is placed at input port of the system.	It is placed at output port of the system
It is used to measure the physical quantity	It is used to measure the continuous and discrete process parameters
It gives information to the system about environment	It accepts command from the system to perform a function.
Example: Photo-voltaic cell which converts light energy into electrical energy.	Example: Stepper motor where electrical energy drives the motor.

12. Light Emitting Diode (LED)

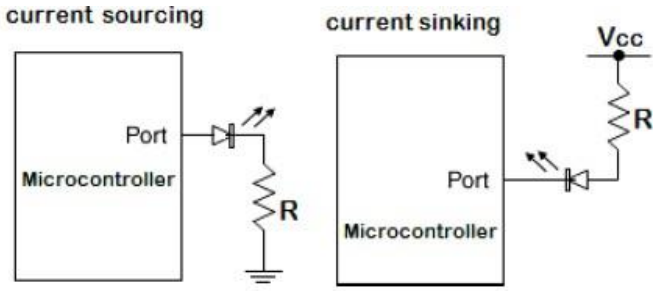
- Light Emitting Diode (LED) is an important output device
- Light Emitting Diode is a p-n junction diode and it contains an anode and a cathode.
- For proper functioning of the LED, the anode of it should be connected to +ve terminal of the supply voltage and cathode to the -ve terminal of supply voltage.
- The current flowing through the LED must be limited to a value below the maximum current that it can conduct.

- A resister is used in series between the power supply and the LED GND to limit the current through the LED. The ideal LED interfacing circuit is shown in Figure



LED interfacing

- LED can be interfaced to the port pin of microcontroller in two methods:
- Method: 1- (current sourcing) Anode of LED is connected to port pin. Cathode is connected to ground (0V) through resistor. When port pin of microprocessor goes logic 1, the LED is forward biased and emits light. When the port pin goes 0, LED is off. That means port pin sources current to LED.
- Method: 2- (current sinking) Cathode of LED is connected to port pin. Anode is connected to external supply through resistor. LED turns on when the port pin is at logic 0. Here port pin sinks current, such that the brightness of LED can be increased to the required level. See fig.



LED interfacing to the port pin of microcontroller

Advantage of LED

- Lower energy consumption
- Longer life time
- Improved physical robustness
- Smaller size
- Faster switching

Application of LED

- Automotive head lamps
- Traffic signals
- General Lighting

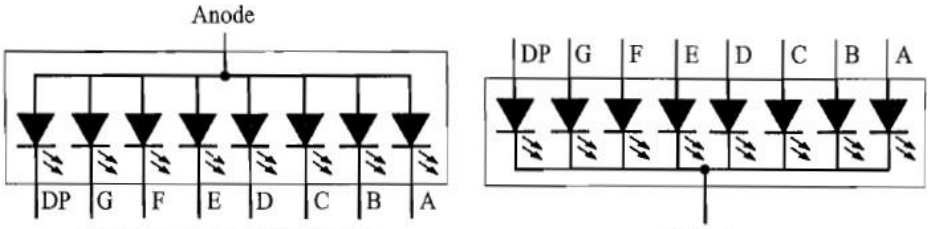
13. 7-Segment LED Display



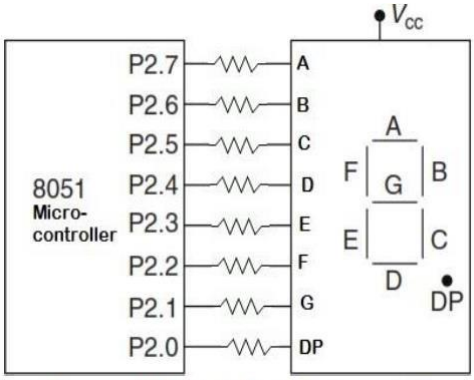
- The 7-segment LED display is an output device for displaying alphanumeric characters.
- It contains 8 LED segments arranged in a special form.
- Out of the 8 LED segments, 7 are used for displaying alpha numeric characters and 1 is used for representing 'decimal point' in decimal number display.

Figure (a) explains the arrangement of LED segments in a 7-segment LED display.

- The LED segments are named A to G and the decimal point LED segment is named as DP.
- The 7-segment LED displays are available in two different configurations, namely; Common Anode and Common Cathode.
- In the common anode configuration, the anodes of the 8 segments are connected commonly whereas in the common cathode configuration, the 8 LED segments share a common cathode line.
- Figure (b) illustrates the Common Anode and Cathode configurations



- Based on the configuration of the 7 – segment LED unit, the LED segment anode or cathode is connected to the port of microcontroller in the order. A segment to the least significant port pin and DP segment to the most significant port pin or vice versa. The current through each segment can be limited by connecting a resistor.



7-segment LED interfacing with microcontroller

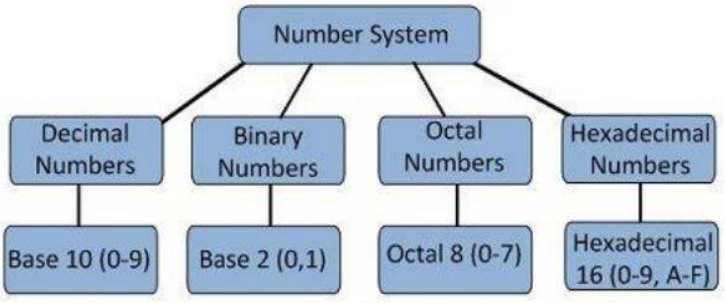
Number	Code	LED status							
		A	B	C	D	E	F	G	DP
0	03	0	0	0	0	0	0	1	1
1	9F	1	0	0	1	1	1	1	1
2	25	0	0	1	0	0	1	0	1
3	0D	0	0	0	0	1	1	0	1
4	D9	1	1	0	1	1	0	0	1
5	49	0	1	0	0	1	0	0	1
6	41	0	1	0	0	0	0	0	1
7	1F	0	0	0	1	1	1	1	1
8	01	0	0	0	0	0	0	0	1
9	19	0	0	0	1	1	0	0	1

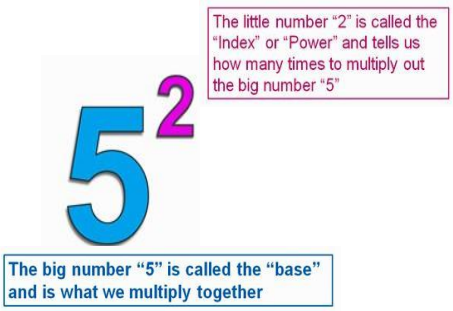


Module-5: Boolean Algebra and Logic Circuits

Number System

- A number system is defined as the representation of numbers by using digits or other symbols in a consistent manner. The value of any digit in a number can be determined by a digit, its position in the number, and the base of the number system. The numbers are represented in a unique manner and allow us to operate arithmetic operations like addition, subtraction, and division.





- In general, a number is expressed in base-r system has coefficients multiplied by powers of r

$$a_n \cdot r^n + a_{n-1} \cdot r^{n-1} + \ldots\ldots\ldots\ldots\ldots\ldots + a_2 \cdot r^2 + a_1 \cdot r^1 + a_0 \cdot r^0 + a_{-1} \cdot r^{-1} + a_{-2} \cdot r^{-2} + \ldots\ldots\ldots\ldots\ldots\ldots + a_{-m} \cdot r^{-m}$$

- For example: 4,265 = 4 × 1,000 + 2 × 100 + 6 × 10 + 5 × 1

1. Decimal number system (Base- 10):

- Decimal number system has only ten (10) digits from 0 to 9. Every number (value) represents with 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9 in this number system. The base of decimal number system is 10, because it has only 10 digits.
- Example: 723_{10}, 32_{10}, 4257_{10}, 4257_{10}

2. Octal number system (Base- 8):

- Octal number system has only eight (8) digits from 0 to 7. Every number (value) represents with 0, 1, 2, 3, 4, 5, 6 and 7 in this number system. The base of octal number system is 8, because it has only 8 digits.
- Example: 35_8, 723_8, 141_8

3. Binary Number System (Base-2):

- A Binary number system has only two digits that are 0 and 1. Every number (value) represents with 0 and 1 in this number system. The base of binary number system is 2, because it has only two digits.
- Example: 10001_2,111101_2,1010101_2

4. Hexadecimal number system (Base- 16):

- A Hexadecimal number system has sixteen (16) alphanumeric values from 0 to 9 and A to F. Every number (value) represents with 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E and F in this number system. The base of hexadecimal number system is 16, because it has 16 alphanumeric values. Here A is 10, B is 11, C is 12, D is 13, E is 14 and F is 15.
- Example: $7B3_{16}$, $6F_{16}$, $4B2A_{16}$

Conversion Rules of Number Systems

1. Conversion of Decimal Number System to Binary / Octal / Hexadecimal Number System

Steps | **Illustration**

1. Identify the base of the required number. ➡ Base is assumed to be 8

2. Divide the given number by the base of the required number and note down the quotient and the remainder in the quotient-remainder form. Repeat this process (dividing the quotient again by the base) until we get the quotient to be less than the base. ➡

8	4320
8	540-0
8	67-4
8	8-3
	1-0

3. The given number in the required number system is obtained just by reading all the remainders and the last quotient from bottom to top. ➡

8	4320
8	540-**0**
8	67- **4**
8	8-**3**
	1-**0**

$\therefore 4320_{10} = 10340_8$

Example:

a. $(401)_{10} = (?)_2$

2	401
2	200-1
2	100-0
2	50-0
2	25-0
2	12-1
2	6-0
2	3-0
	1-1

$(401)_{10} = (110010001)_2$

b. $(401)_{10} = (?)_{16}$

16	401
16	25-1
	1-9

$(401)_{10} = (191)_{16}$

c. $(401)_{10} = (?)_{16}$

16	254
	15-E

$(401)_{10} = (FE)_{16}$

1. a. Conversion of fractional decimal to binary

Steps

1. Multiply fraction by 2 keeping notice of the resulting integer and fractional part.
2. Continue multiplying by 2 until you get a resulting fractional part equal to zero.
3. Write out the integer parts from the results of each multiplication.

Illustration

Conversion of .47 to binary

0.47 * 2 = 0.94, Integral part: 0
0.94 * 2 = 1.88, Integral part: 1
0.88 * 2 = 1.76, Integral part: 1

So equivalent binary of fractional part of decimal is .011

Example:

a. Convert Decimal 126.25 to Binary

Conversion of 126 to binary

2	126
2	63-0
2	31-1
2	15-1
2	7-1
2	3-1
	1-1

$(11111110)_2$

Conversion of .25 to binary

1. 0.25 * 2 = 0.50, Integral part: 0
2. 0.50 * 2 = 1.00, Integral part: 1

$(.01)_2$

$(126.25)_{10}$ =$(1111110.01)_2$

2. Conversion of binary to decimal/octal/hexadecimal

2. a. Binary to octal

1. Partition the binary number into groups of *three* digits, starting from the binary point & proceeding to left & right.
2. Assign octal digits to each group.

Example:

a. Convert $(10110001101011.111100000110)_2$ to Octal

10 110 001 101 011 . 111 100 000 110

2 6 1 5 3 7 4 0 6

$(26153.7406)_8$

b. Convert $(110110.11)_2$ to Octal

110 110 . 110

6 6 6

$(66.6)_8$

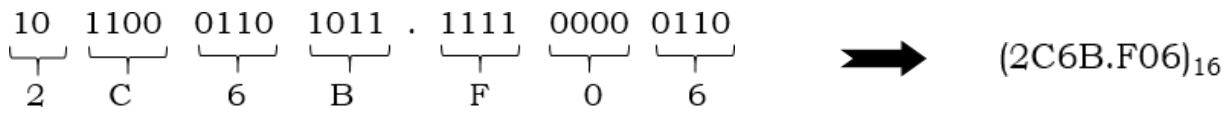
2. b. Binary to Hexadecimal

Steps:

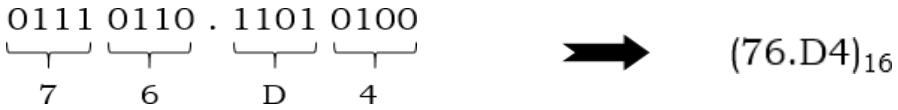
1. Partition the binary number into groups of four digits, starting from the binary point & proceeding to left & right.
2. Assign hexadecimal digits to each group.

Example:

a. Convert $(10110001101011.111100000110)_2$ to hexadecimal



b. Convert Binary 1110110.110101 to hexadecimal



2. c. Binary to Decimal

Steps:

1. List out the powers of 2 for all the digits starting from the rightmost position.
2. Now multiply each digit in the binary number starting from the right with its respective weight based on its position and evaluate the product.
3. Finally, sum up all the products obtained for all the digits in the binary number.

Example

a. Convert $(10011011.110)_2 = (?)_{10}$

1	0	0	1	1	0	1	1	.	1	1	0
↓	↓	↓	↓	↓	↓	↓	↓		↓	↓	↓
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	.	2^{-1}	2^{-2}	2^{-3}
↓	↓	↓	↓	↓	↓	↓	↓		↓	↓	↓
2^7	0	0	2^4	2^3	0	2^1	2^0	.	2^{-1}	2^{-2}	0
↓	↓	↓	↓	↓	↓	↓	↓		↓	↓	↓
128	0	0	16	8	0	2	1	.	0.5	0.25	0

↓

155.75_{10}

$\therefore 10011011.110_2 = 155.75_{10}$

3. Conversion of hexadecimal to binary/decimal/octal

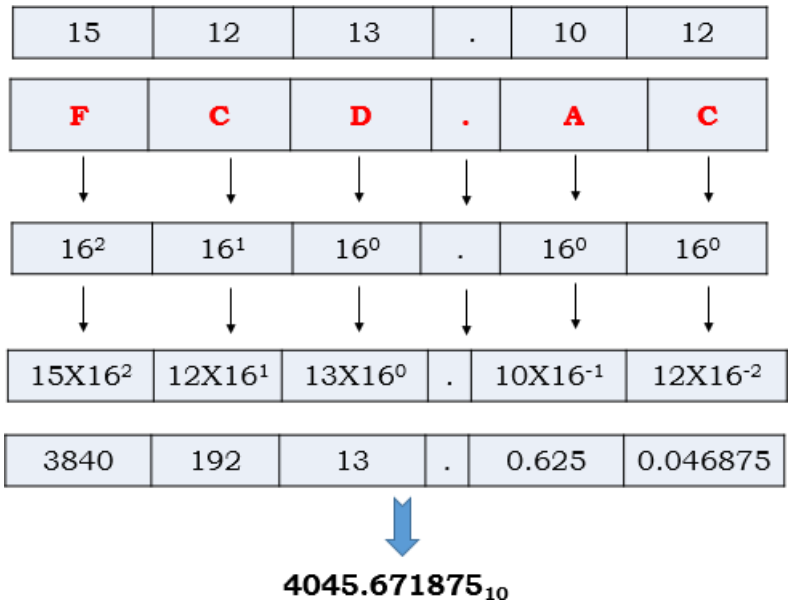
3. a. Hexadecimal to Decimal

Steps:

1. List out the powers of 16 for all the digits starting from the rightmost position.
2. Now multiply each digit in the binary number starting from the right with its respective weight based on its position and evaluate the product.
3. Finally, sum up all the products obtained for all the digits in the binary number.

Example:

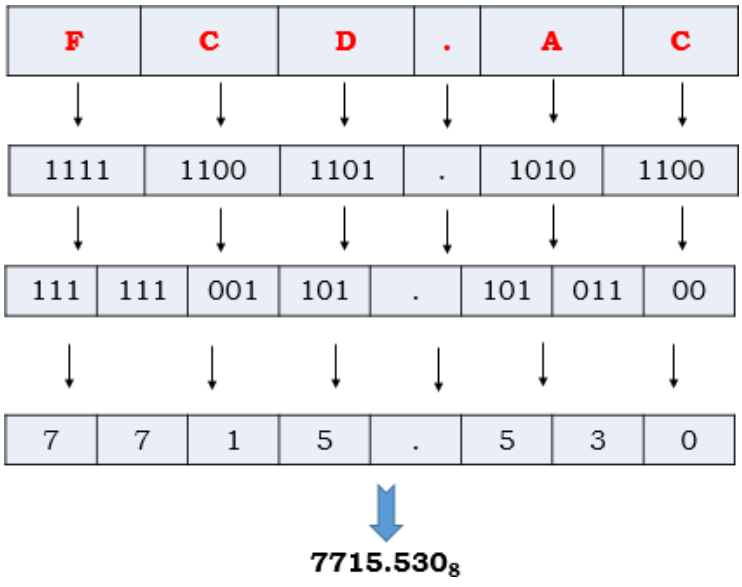
a. Convert $(FCD.AC)_{16} = (?)_{10}$



3. b. Hexadecimal to Octal

Steps:

1. Represent hexadecimal digits to their binary equivalent.
2. Partition the binary number into groups of three digits, starting from the binary point & proceeding to left & right.
3. Assign octal digits to each group.



3. c. Hexadecimal to Binary:

Steps: Refer table

4. Conversion of Octal to decimal/hexadecimal/binary

4.a. Octal to Decimal:

Steps:

1. List out the powers of 8 for all the digits starting from the rightmost position.
2. Now multiply each digit in the octal number starting from the right with its respective weight based on its position and evaluate the product.
3. Finally, sum up all the products obtained for all the digits in the binary number.

4. b. Octal to Hexadecimal:

Steps:

1. Convert each octal digit to its binary (refer table) then group binary digits in a group of 4 and use table to represent hexadecimal digits for each group.

4. c. Octal to Binary:

Steps: Refer table

Table

Dec	Hex	Oct	Bin
0	0	000	0000
1	1	001	0001
2	2	002	0010
3	3	003	0011
4	4	004	0100
5	5	005	0101
6	6	006	0110
7	7	007	0111
8	8	010	1000
9	9	011	1001
10	A	012	1010
11	B	013	1011
12	C	014	1100
13	D	015	1101
14	E	016	1110
15	F	017	1111

Binary Addition

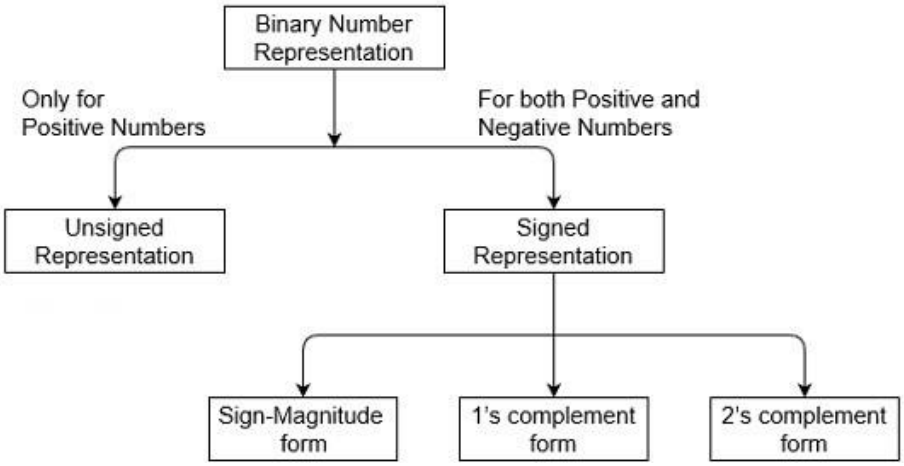
- When adding binary numbers, there are four points or steps to remember before proceeding through the operation. These are:

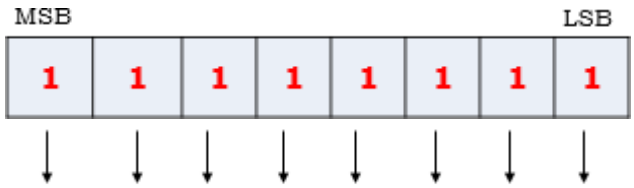
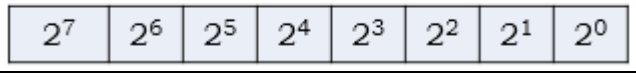
$$0+0=0$$
$$0+1=1$$
$$1+0=1$$
$$1+1=0\ (carry\ 1\ to\ the\ next\ significant\ bit)$$

- As there no numbers other than 0 and 1 in the binary number system these four steps include all the possible operations of addition.

Carry →	1	1		1	1		1	
Number-1 →	1	0	1	0	1	1	0	1
Number-2 →	0	1	1	0	1	1	0	1
	1	0	0	0	1	1	0	1	0

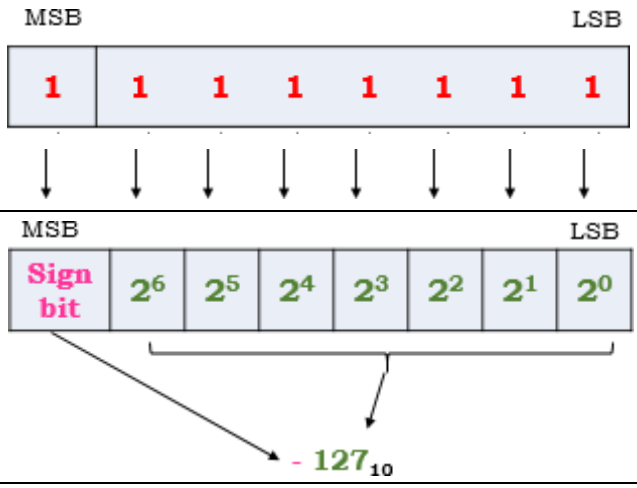
Binary signed /unsigned number representation



An 8-bit Unsigned binary number	
Powers of 2 representation	
Adding all powers of 2 gives	255_{10}
∴ ***Range of an 8-bit unsigned number is 0-255 (0-2^8-1), in general 0- (2^N-1)***	

Sign magnitude representation

- The sign-magnitude binary format is the simplest conceptual format. In this method of representing signed numbers, the most significant bit (MSB) takes on extra meaning.
  - If the MSB is a 0, this indicates the number as a positive.
  - If the MSB is a 1, this indicates that the number is negative.
  - The other bits indicate the magnitude (absolute value) of the number.

<table>
<tr><td>Consider an 8-bit number signed binary no</td><td rowspan="2"> 
</td></tr>
<tr><td>Powers of 2 representation</td></tr>
<tr><td>Adding all powers of 2 gives</td><td>
</td></tr>
<tr><td colspan="2">∴ Range of an 8-bit signed number is -127 to +127, in general $-(2^{N-1})$ to $+(2^{N-1})$</td></tr>
</table>

Representation of negative numbers

1. **Sign-Magnitude form:**
   - For n bit binary number, 1 bit is reserved for sign symbol.
   - If the value of sign bit is 0, then the given number will be positive, else if the value of sign bit is 1, then the given number will be negative. Remaining (n-1) bits represent magnitude of the number.
   - *Drawback: Since magnitude of number zero (0) is always 0, so there can be two representation of number zero (0), positive (+0) and negative (-0), which depends on value of sign bit. Hence these representations are ambiguous generally because of two representation of number zero (0).*

2. **1's complement form:**
   - In signed number there is extra bit for sign representation. If value of sign bit is 0, then number is positive and can be represented directly in simple binary form, but if value of sign bit is 1, then number is negative and it is to be represented in 1's complement of given binary number.
   - 1's complement of a number is obtained by inverting each bit of given number.
   - *Drawback: One gets negative number by 1's complement of a positive number and positive number by using 1's complement of a negative number. Therefore, in this representation, zero (0) can have two representation, that's why 1's complement form is also ambiguous form*

3. **2's complement form:**
   - Since, 2's complement of a number is obtained by inverting each bit of given number plus 1 to least significant bit (LSB).
   - The algorithm to subtract two binary number using 2's complement is explained as following below −
     1. Take 2's complement of the subtrahend
     2. Add with minuend
     3. If the result of above addition has carry bit 1, then it is dropped and this result will be positive number.
     4. If there is no carry bit 1, then take 2's complement of the result which will be negative.

Example

a. Perform subtraction using 2's complement: **$(11010)_2$-$(10000)_2$**

Steps:

1. Take 2's complement of the subtrahend
2. Add with minuend
3. If the result of above addition has carry bit 1, then it is dropped and this result will be positive number.
4. If there is no carry bit 1, then take 2's complement of the result which will be negative

2's Complement computation

1's Complement of Subtrahend =>	0	1	1	1	1
Adding 1 to 1's complement =>					1
2's complement subtrahend=>	1	0	0	0	0

Minuend =>		1	1	0	1	0
2's complement of subtrahend =>		1	0	0	0	0
Discard carry generated =>	1	0	1	0	1	0
Result =>		**0**	**1**	**0**	**1**	**0**

Verification: 26_{10}-16_{10}=10_{10}

a. Perform subtraction using 2's complement: **100_2-110000_2**

2's Complement computation

1s Complement of Subtrahend =>	0	0	1	1	1	1
Adding 1 to 1's complement =>						1
2's complement subtrahend=>	0	1	0	0	0	0

Minuend =>				1	0	0
2's complement subtrahend =>	0	1	0	0	0	0
Result without carry =>	**0**	**1**	**0**	**1**	**0**	**0**
1's complement of Result =>	1	0	1	0	1	1
Adding 1 to 1's complement =>						1
Result, which is a negative no.=>	**1**	**0**	**1**	**1**	**0**	**0**

Verification: 4_{10} - 48_{10}= - 44_{10}

Note:

1. If either one of the given two numbers is –ve, then the above listed procedure is to be followed.
2. If both the given numbers are –ve then,
   a. Take 2s complement for both the numbers with bit length equaling the results bit length.
   b. Add both the 2s complement versions of two –ve numbers.
   c. Since a carry is generated out of MSB, this carry has to be disregarded & the result is in 2s complement indicating a –ve result.

Basic Logic Gates

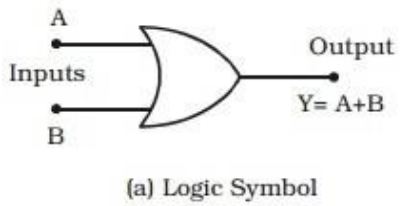
- A logic gate is a basic building block of a digital circuit that has two inputs and one output. The relationship between the i/p and the o/p is based on a certain logic. These gates are implemented using electronic switches like transistors, diodes.
- In practice, basic logic gates are built using CMOS technology, FETS, and MOSFET (Metal Oxide Semiconductor FET).
- Logic gates are used in microprocessors, microcontrollers, embedded system applications, and in electronic and electrical project circuits.
- The basic logic gates are categorized into seven: AND, OR, XOR, NAND, NOR, XNOR, and NOT.

Truth Tables

- Truth tables are used to show the function of a logic gate. All the possible inputs and outputs of a logic circuit are represented in a table called *Truth Table*.

1. OR Gate

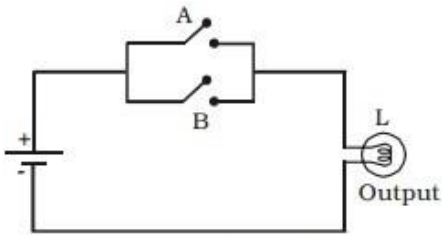
- An OR gate has two or more inputs but only one output. It is known as OR gate, because the output is high if any one or all of the inputs are high. The logic symbol of a two input OR gate is shown in Fig a.
- The Boolean expression to represent OR gate is given by Y=A+B ('+'symbol should be read as OR)
- The OR gate can be thought of like an electrical circuit shown in Fig b, in which switches are connected in parallel with each other. The lamp will glow if either the inputs are closed or any one of them is closed. The operation of OR gate is shown in Truth Table



(a) Logic Symbol

Table Truth table of OR gate

Inputs		Output
A	B	Y = A + B
0	0	0
0	1	1
1	0	1
1	1	1

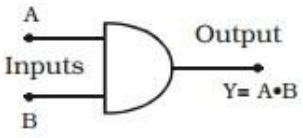


(b)Electrical circuit

2. AND Gate

- An AND gate has two or more inputs but only one output. It is known as AND gate because the output is high only when all the inputs are high. The logic symbol of a two input AND gate is shown in Fig a
- The Boolean expression to represent OR gate is given by Y=A • B ('•'symbol should be read as AND)

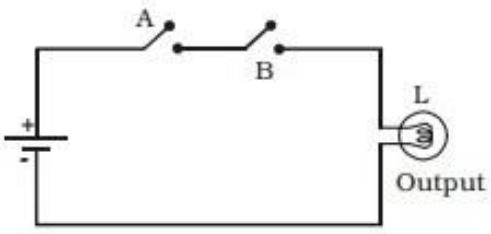
- The AND gate can be thought of like an electrical circuit shown in Fig b, in which switches are connected in series with each other. The lamp will glow if both the inputs are closed. The operation of AND gate is shown in Truth Table.



(a) Logic symbol

Inputs		Output
A	B	Y = A · B
0	0	0
0	1	0
1	0	0
1	1	1

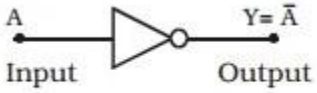
Table Truth table of AND gate



(b) Electrical Circuit

3. NOT Gate

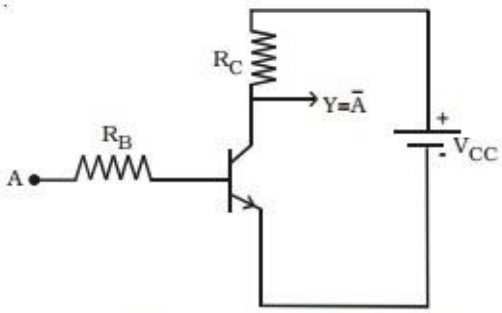
- The NOT gate is a gate with only one input and one output. It is so called, because its output is complement to the input. It is also known as inverter. Fig a shows the logic symbol for NOT gate
- The Boolean expression to represent NOT gate is given by **Y= $\bar{A}$** (‘ ¯ ’symbol should be read as complement of the symbol under it)
- The NOT gate can be thought of like an electrical circuit as shown in Fig b. When switch A is closed, input is high and the bulb will not glow (i.e) the output is low and vice versa. The operation of NOT gate is shown in Truth Table.



(a) logic symbol

Input	Output
A	Y = $\bar{A}$
0	1
1	0

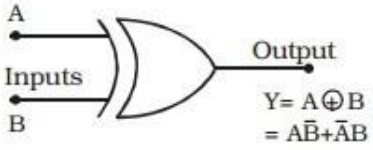
Truth Table of NOT gate



NOT gate using transistor

4. Exclusive OR gate (EXOR gate)

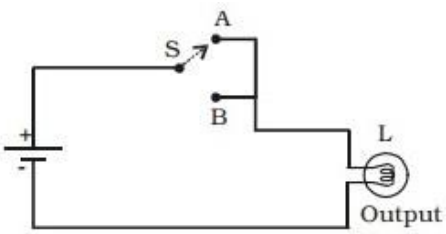
- The logic symbol for exclusive OR (EXOR) gate is shown in fig.a
- The Boolean expression to represent EX-OR operation is Y = A ⊕ B
- EXOR gate has an output 1, only when the inputs are complement to each other. The equivalent switching circuit is shown in Fig b.
- Switch positions A and B will individually make the lamp to be ON. But the combination of A and B is not possible. The EXOR operation is represented in Table.



(a) Logic symbol

Inputs		Output
A	B	Y = A ⊕ B
0	0	0
0	1	1
1	0	1
1	1	0

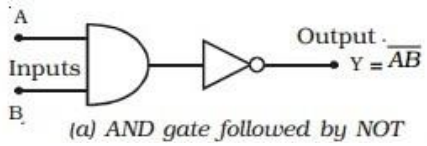
Truth table of EXOR gate



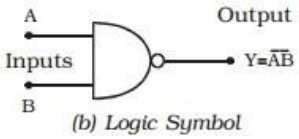
(b) Electrical circuit

5. NAND gate

- This is a NOT-OR gate. It can be made out of an OR gate by connecting an inverter at its output (Fig a).
- NAND gate function is reverse of AND gate function. A NAND gate will have an output, only if both inputs are not 1. In other words, it gives an output 1, if either A or B or both are 0. The operation of a NAND gate is represented in Truth Table.
- The Boolean expression to represent NAND operation is $Y = \overline{A \cdot B}$



(a) AND gate followed by NOT



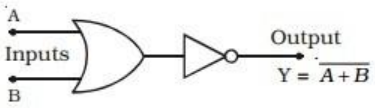
(b) Logic Symbol

Inputs		Output Y = $\overline{AB}$
A	B	
0	0	1
0	1	1
1	0	1
1	1	0

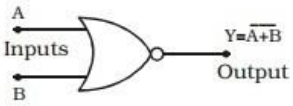
Truth table of NAND gate

6. NOR gate

- This is a NOT-OR gate. It can be made out of an OR gate by connecting an inverter at its output (Fig a). The logic symbol for NOR gate is given in Fig b
- The NOR gate function is the reverse of OR gate function. A NOR gate will have an output, only when all inputs are 0. In a NOR gate, output is high, only when all inputs are low. The NOR operation is represented in Table.
- The Boolean expression to represent XNOR operation is $Y = \overline{A + B}$



(a) OR gate followed by NOT

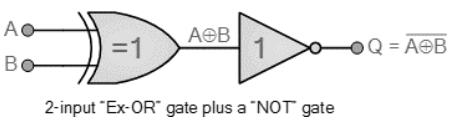


Inputs		Output Y = $\overline{A+B}$
A	B	
0	0	1
0	1	0
1	0	0
1	1	0

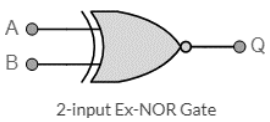
Truth table of NOR gate

7. XNOR gate

- The XNOR gate (also known as a XORN'T, ENOR, EXNOR or NXOR) – and pronounced as Exclusive NOR – is a digital logic gate whose function is the logical complement of the exclusive OR gate (XOR gate). Logically, an XNOR gate is a NOT gate followed by an XOR gate.
- In the XOR gate operation, the output is only 1 when only one input is 1. The output is logical 0 when both inputs are the same, meaning they are either 1 or 0. But in the XNOR gate, the inverse is true. Hence the output is 0 when only one input is 0, and the output is 1 when both inputs are the same (i.e. either two 0's or two 1's).
- The Boolean expression to represent XNOR operation is $Y = \overline{A \oplus B}$



2-input "Ex-OR" gate plus a "NOT" gate



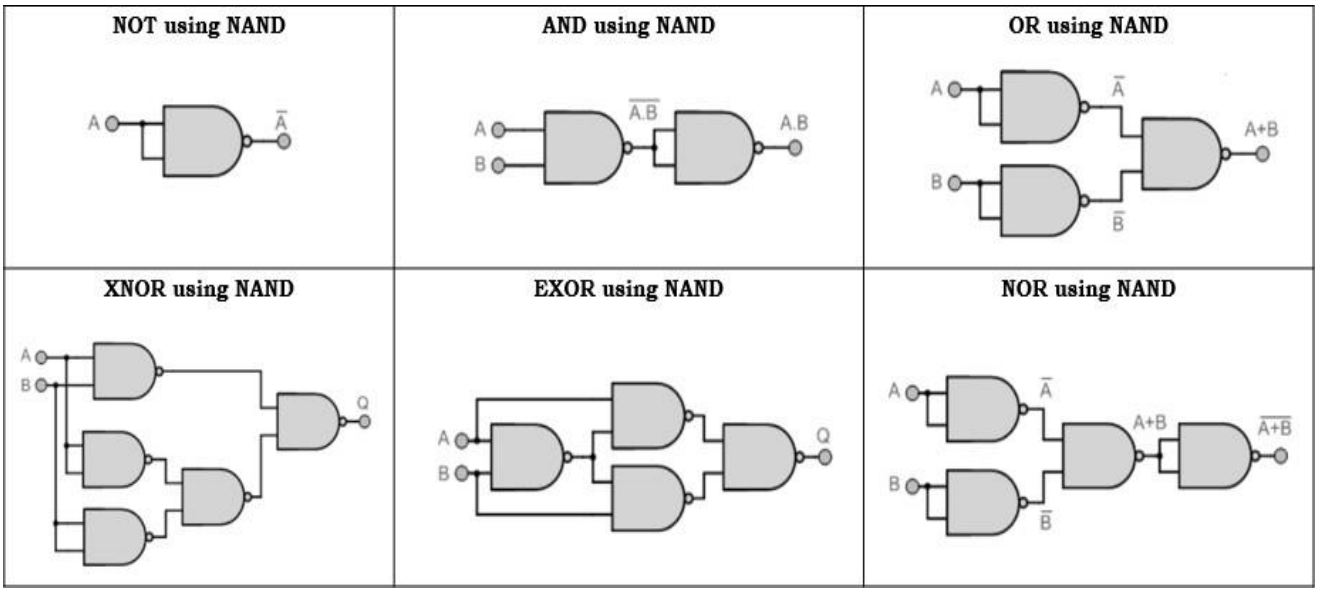
2-input Ex-NOR Gate

A	B	Q
0	0	1
0	1	0
1	0	0
1	1	1

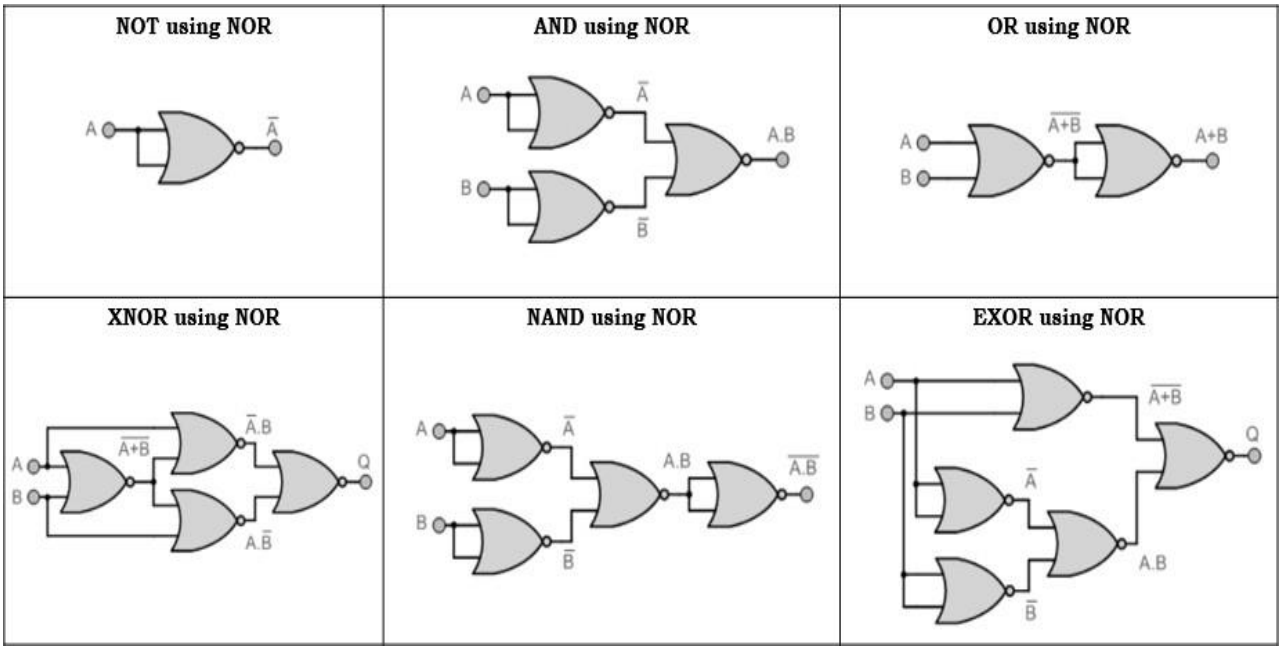
Universal Gates:

- A universal gate is a logic gate which can implement any Boolean function without the need to use any other type of logic gate. The NOR gate and NAND gate are universal gates. This means that you can create any logical Boolean expression using only NOR gates or only NAND gates.

1. NAND Gate as Universal Gate:



2. NOR Gate as Universal Gate



Canonical Forms:

- Canonical is a word used to describe a 'condition of a switching equations'. There are two types of canonical forms:
    1. Disjunctive Normal Forms or Sum of Products or (SOP).
    2. Conjunctive Normal Forms or Products of Sums or (POS).

1. Canonical Minterm Form:

- If each term in SOP form contains all the literals, then the SOP form is called as Standard SOP form or Canonical SOP Form. Each individual term in the standard SOP form is called Minterm. Therefore, canonical SOP form is also known as Canonical Minterm.
- A minterm, denoted as m_i, where $0 \leq i < 2^n$, is a product (AND) of the n variables in which each variable is complemented if the value assigned to it is 0, and uncomplemented if it is 1.
    - 1-minterms = minterms for which the function F = 1.
    - 0-minterms = minterms for which the function F = 0

1.1. Σ- notation:

- The Σ-notation is used to simplify writing functions in Canonical Minterm Forms. Minterms are represented by the row numbers for the function that evaluates to 1. Such Σ-notations is also referred as 'decimal formats'.
- Any Boolean function can be expressed as a sum (OR) of its 1- minterms. A shorthand notation:

F(list of variables) = Σ(list of 1-minterm indices)

1.2. Conversion to Standard SOP form:

- If the given SOP equation is not in the standard SOP form then it should be converted to standard form by following below steps:
    1. Find the missing literal in each product term
    2. AND each product terms having missing literal/s with terms/s formed by ORing the literal & its compliment. By this the original AND term is value is not changed.
    3. Expand the terms by applying distributive law & reorder the literals in the product terms
    4. Reduce the expression by omitting repeated product terms if any.
- Consider the table is shown below:

Row No	a	b	c	f	Minterms
0	0	0	0	0	$m_0 = \bar{a}\,\bar{b}\,\bar{c}$
1	0	0	1	0	$m_1 = \bar{a}\,\bar{b}\,c$
2	0	1	0	**1**	$\mathbf{m_2 = \bar{a}\,b\,\bar{c}}$
3	0	1	1	**1**	$\mathbf{m_3 = \bar{a}\,b\,c}$
4	1	0	0	0	$m_4 = a\,\bar{b}\,\bar{c}$
5	1	0	1	**1**	$\mathbf{m_5 = a\,\bar{b}\,c}$
6	1	1	0	0	$m_6 = a\,b\,\bar{c}$
7	1	1	1	0	$m_7 = a\,b\,c$

- From above table function is $f = \bar{a}\,b\,\bar{c} + \bar{a}\,b\,c + a\,\bar{b}\,c$ and is expressed in the canonical minterm form in Σ-notation as $f = m_2 + m_3 + m_5 = \Sigma(m2, m3, m5) = \Sigma m(2, 3, 5)$.

2. Canonical Maxterm Form:

- If each term in POS form contains all the literals, then the POS form is called as Standard POS form or Canonical POS Form. Each individual term in the standard POS form is called Maxterm. Therefore, canonical POS form is also known as Canonical Maxterm Form.
- A maxterm, denoted as Mi, where $0 \leq i < 2^n$, is a sum (OR) of the n variables (literals) in which each variable is complemented if the value assigned to it is 1, and uncomplemented if it is 0.
  - 1-maxterms = maxterms for which the function F = 0.
  - 0-maxterms = maxterms for which the function F = 1.

2.1. Π-notation:

- The Π-notation is used to simplify writing in canonical maxterm form. Maxterms are represented by Π along with row numbers for which the function evaluates to 0. Such Π-notation is also referred as 'decimal formats'.
- Any Boolean function can be expressed as a product (AND) of its 0-maxterms. A shorthand notation: *F(list of variables) = Π(list of 0-maxterm indices)*

2.2. Conversion to Standard POS form:

- If the given POS equation is not in the standard POS form then it should be converted to standard form by following below steps:
  1. Find the missing literal in each sum term
  2. OR each sum terms having missing literal/s with terms/s formed by ANDing the literal & its compliment. By this the original OR term value is not changed.
  3. Expand the terms by applying distributive law & reorder the literals in the sum terms
  4. Reduce the expression by omitting repeated sum terms if any.

- Consider the table of a function as shown below:

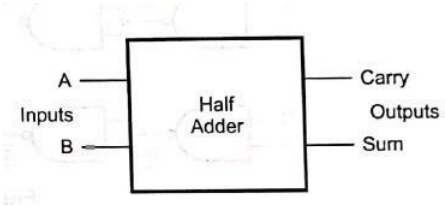
Row No	a	b	c	f	Maxterms
0	0	0	0	**0**	$\mathbf{M_0 = a + b + c}$
1	0	0	1	1	$M_1 = a + b + \bar{c}$
2	0	1	0	1	$M_2 = a + \bar{b} + c$
3	0	1	1	**0**	$\mathbf{M_3 = a + \bar{b} + \bar{c}}$
4	1	0	0	**0**	$\mathbf{M_4 = \bar{a} + b + c}$
5	1	0	1	**0**	$\mathbf{M_5 = \bar{a} + b + \bar{c}}$
6	1	1	0	1	$M_6 = \bar{a} + \bar{b} + c$
7	1	1	1	1	$M_7 = \bar{a} + \bar{b} + \bar{c}$

- From above table function is $f = (a + b + c)(a + \bar{b} + \bar{c})(\bar{a} + b + c)(\bar{a} + b + \bar{c})$ and is expressed in the canonical maxterm form in Π -notation as

$$f = M_0 + M_3 + M_4 + M_5 = \Pi(M_0 + M_3 + M_5 + M_5) = \Pi M(0, 3, 4, 5)$$

Half Adder

- A half adder is a logical circuit that performs an addition operation on two binary digits. The half adder produces a sum and a carry value which are both binary digits.



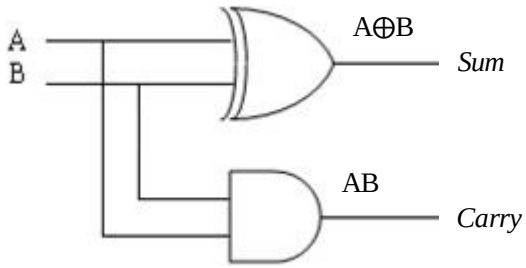
Truth Table

Inputs		Outputs	
A	B	Sum S	Carry C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Boolean expression for Sum & Carry outputs:

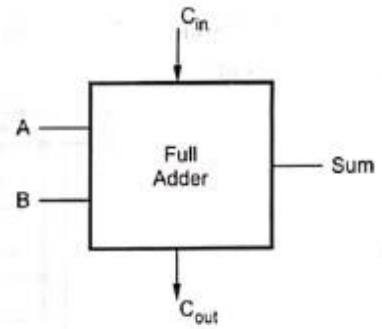
$Sum, S = \bar{A}B + A\bar{B}$ $\boldsymbol{Sum, S = A \oplus B}$	$\boldsymbol{Carry, C = AB}$

Implementation using basic gates:



Full Adder

- A full adder has three inputs. The two single bit data inputs A and B as in Half Adder and an additional Carry-in (Cin) input to receive the carry from a previous stage as shown below.



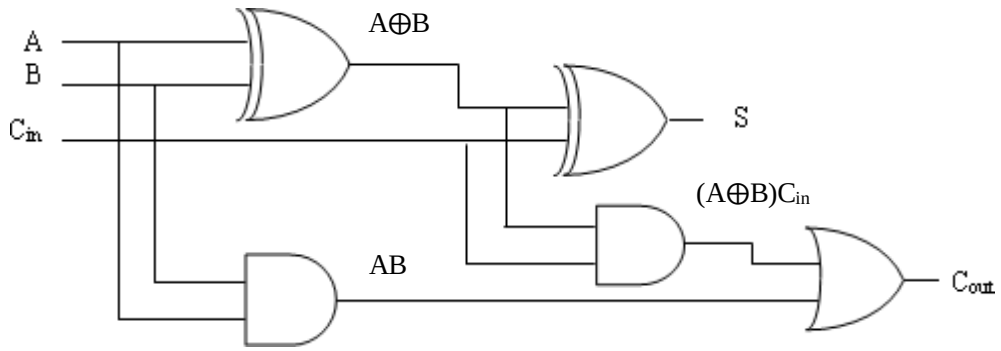
Truth Table

Inputs			Outputs	
A	B	C_{in}	Sum S	Carry C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Boolean expression for Sum & Carry outputs:

$S = \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in} + ABC_{in}$ $S = C_{in}(AB + \bar{A}\bar{B}) + \bar{C}_{in}(\bar{A}B + A\bar{B})$ $S = C_{in}(\bar{A} \oplus \bar{B}) + \bar{C}_{in}(A \oplus B)$ $\boldsymbol{S = A \oplus B \oplus C_{in}}$	$C_{out} = \bar{A}BC_{in} + A\bar{B}C_{in} + AB\bar{C}_{in} + ABC_{in}$ $C_{out} = C_{in}(\bar{A}B + A\bar{B}) + AB(C_{in} + C_{in})$ $\boldsymbol{C_{out} = C_{in}(A \oplus B) + AB}$

Full Adder Implementation using EXOR and basic gates:



Inspire: “ Success is a journey, not a Destination.”